Pattern Recognition & Machine Learning

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Required textbook:

李航，统计学习，清华大学出版社

周志华，机器学习，清华大学出版社

Christopher M.Bishiop, patter recognition and machine learning,

- Pattern recognition has its origins in engineering, whereas machine learning grew out of computer science.
- However, these activities can be viewed as two facets of the same field, and together they have undergone substantial development over the past ten years. （Christopher M.Bishiop， 2006）

机器学习

机器学习是从人工智能中产生的一个重要学科分支，是实现智能化的关键

经典定义：利用经验改善系统自身的性能

经验 → 数据

随着该领域的发展，目前主要研究智能数据分析的理论和算法，并已成为智能数据分析技术的源泉之一

图灵奖连续授予在该方面取得突出成就的学者

Leslie Valiant
(1949 -)
(Harvard Univ.)
“计算学习理论”奠基人

2011 年度

Judea Pearl
(1936 -)
(UCLA)
“图模型学习方法”先驱

2012 年度

机器学习

（Machine Learning）

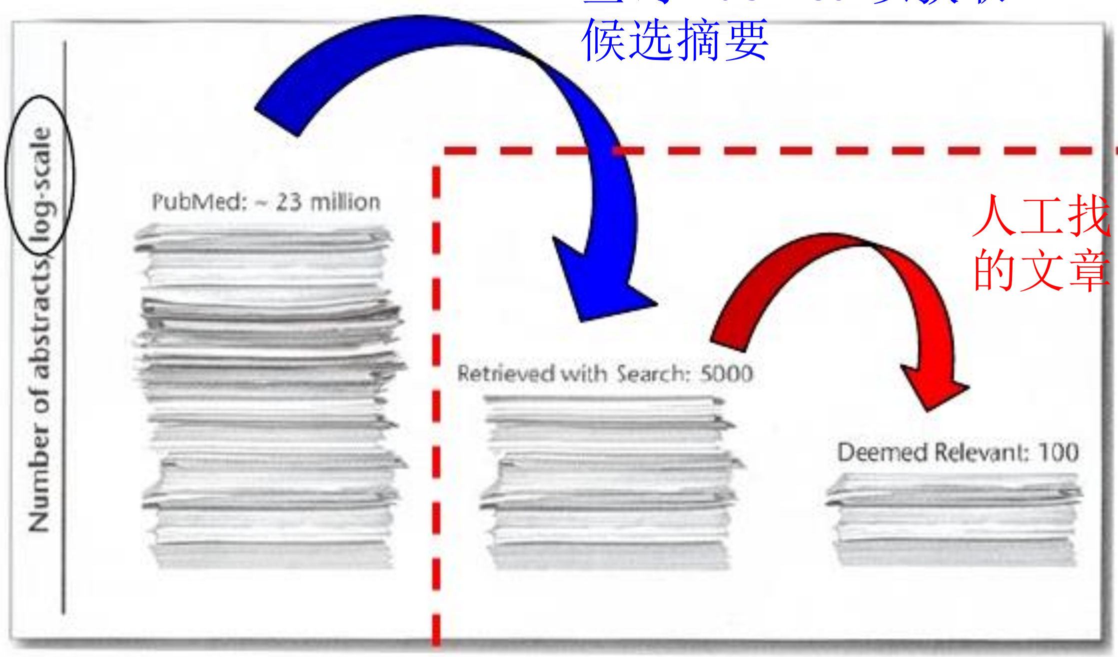
究竟是什么东东？

看个例子

“文献筛选”的故事

在“循证医学”（evidence-based medicine）中，针对特定的临床问题，先要对相关研究报告进行详尽评估

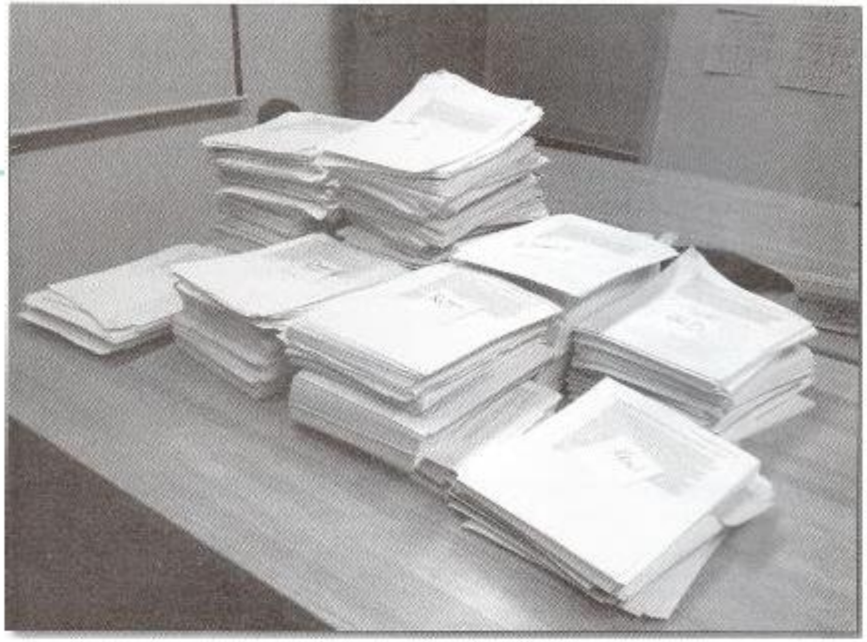
查询 PubMed 以获取候选摘要



人工找出值得全文审读的文章

[C. Brodley et al., AI Magazine 2012]

“文献筛选”的故事



a portion of the 33,000 abstracts

在一项关于婴儿和儿童残疾的研究中，美国Tufts医学中心筛选了约 33,000 篇摘要

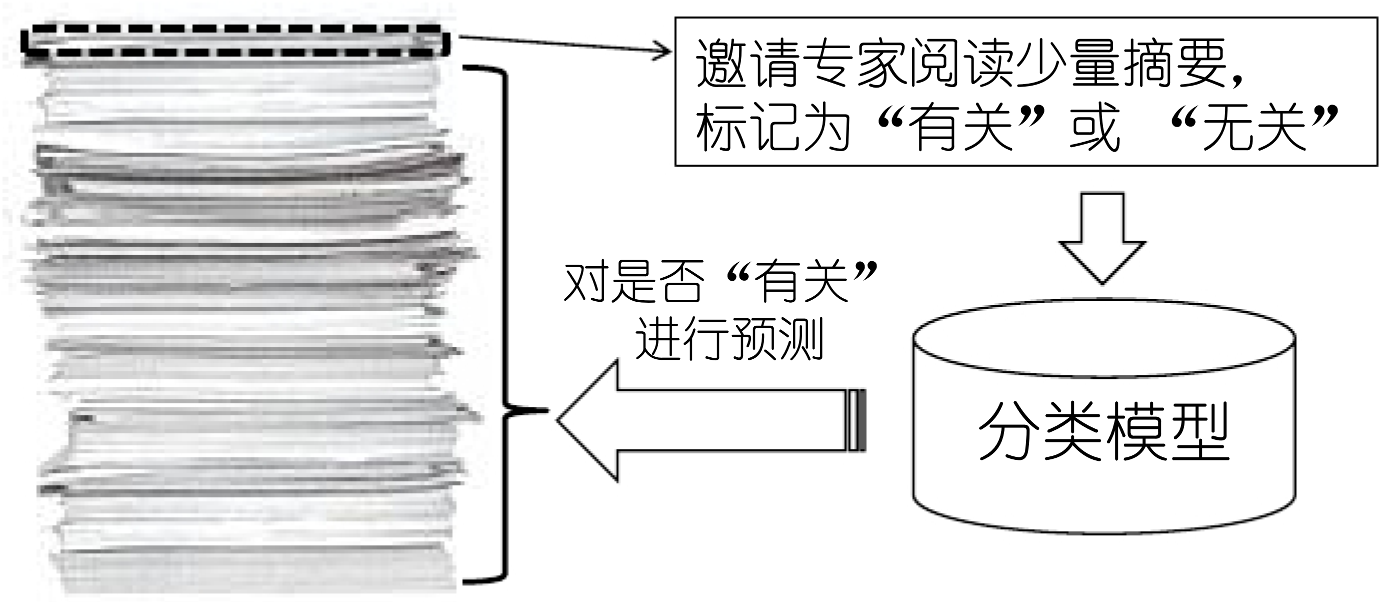
尽管Tufts医学中心的专家效率很高，对每篇摘要只需 30 秒钟，但该工作仍花费了 250 小时

每项新的研究都要重复这个麻烦的过程！

需筛选的文章数在不断显著增长！

“文献筛选” 的故事

为了降低昂贵的成本，Tufts医学中心引入了机器学习技术

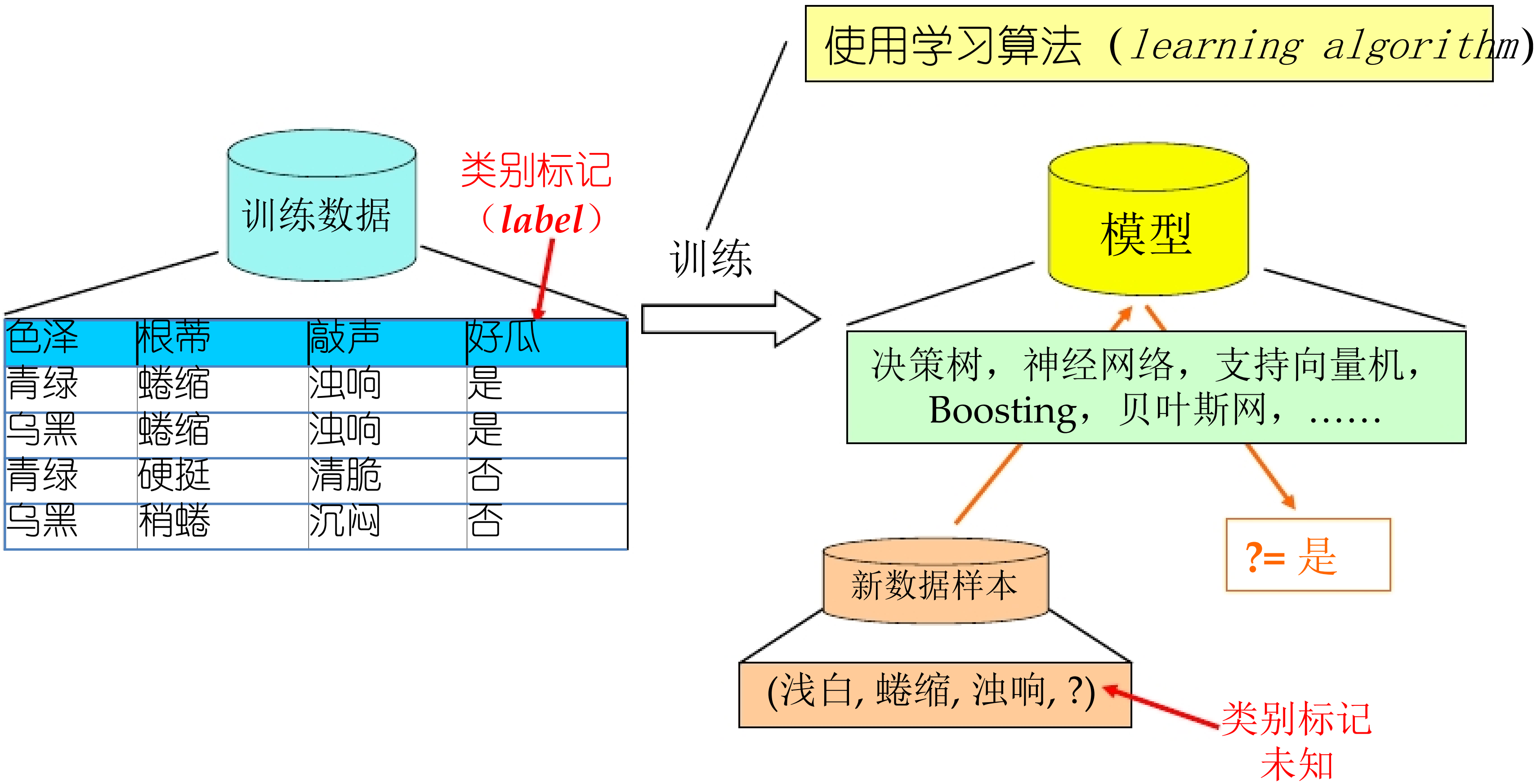


人类专家只需阅读 50 篇摘要，系统的自动筛选精度就达到 93%

人类专家阅读 1,000 篇摘要，则系统的自动筛选敏感度达到 95%

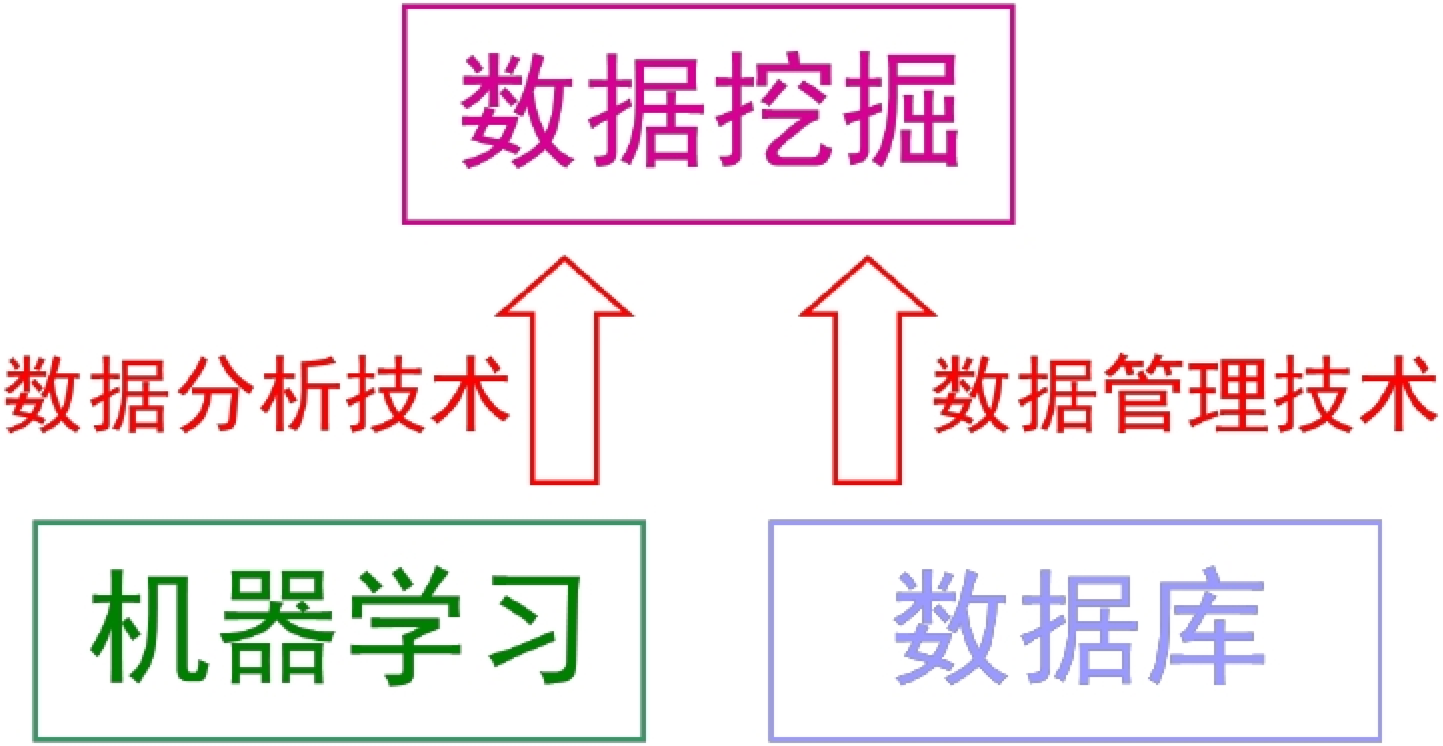
（人类专家以前需阅读 33,000 篇摘要才能获得此效果）

典型的机器学习过程



色泽	根蒂	敲声	好瓜
青绿	蜷缩	浊响	是
乌黑	蜷缩	浊响	是
青绿	硬挺	清脆	否
乌黑	稍蜷	沉闷	否

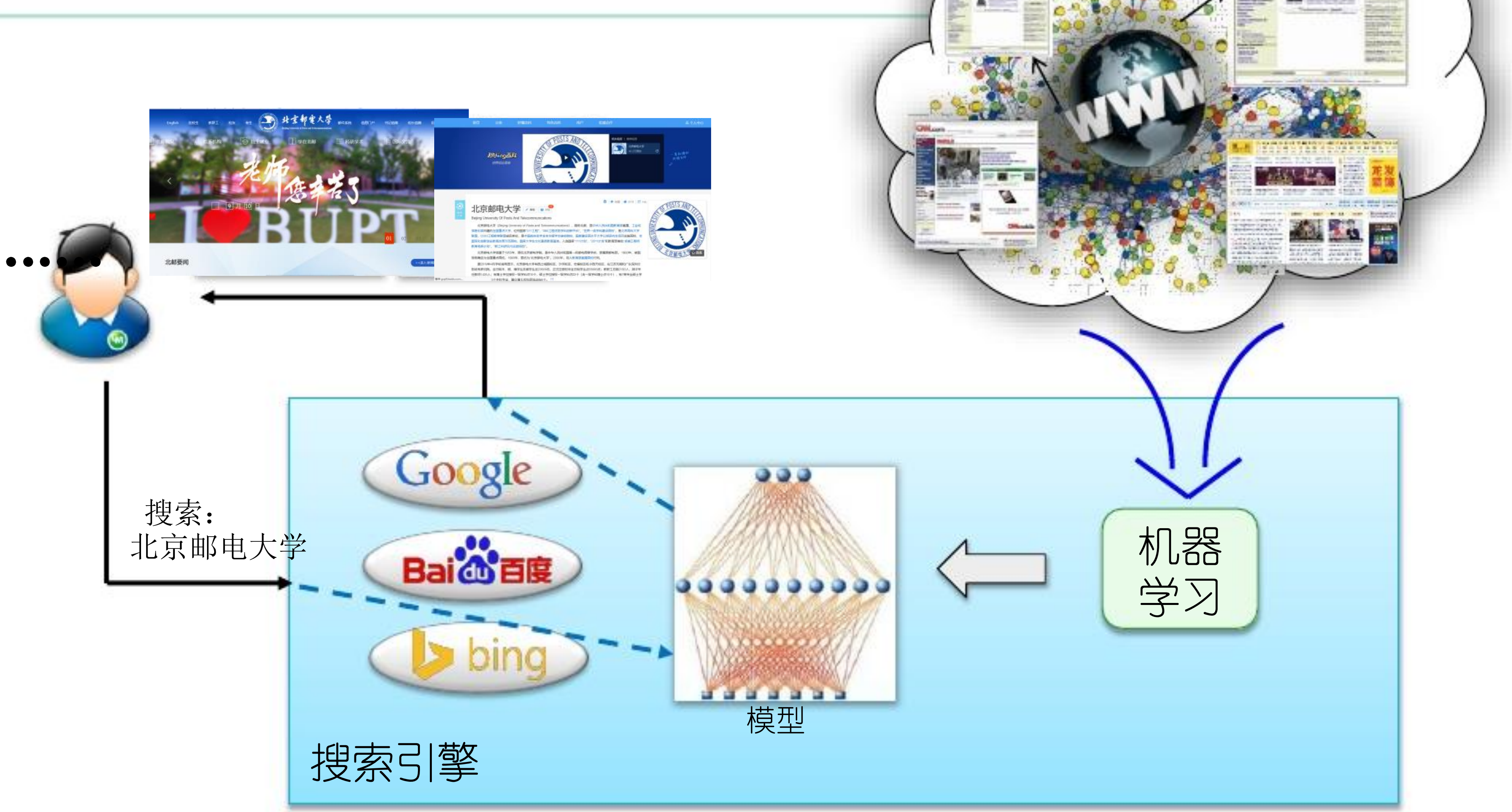
机器学习与数据挖掘



机器学习能做什么？

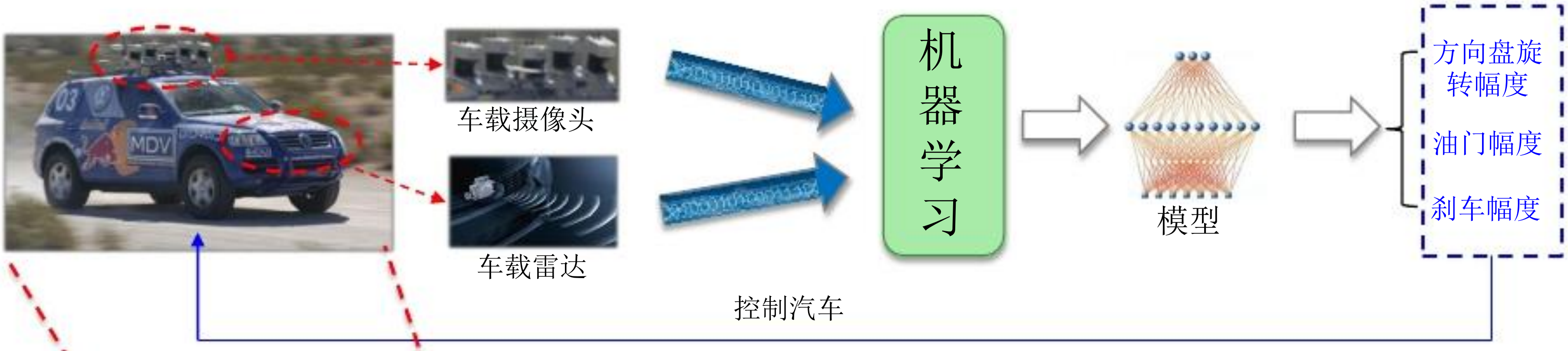
我们可能每天都
在用机器学习

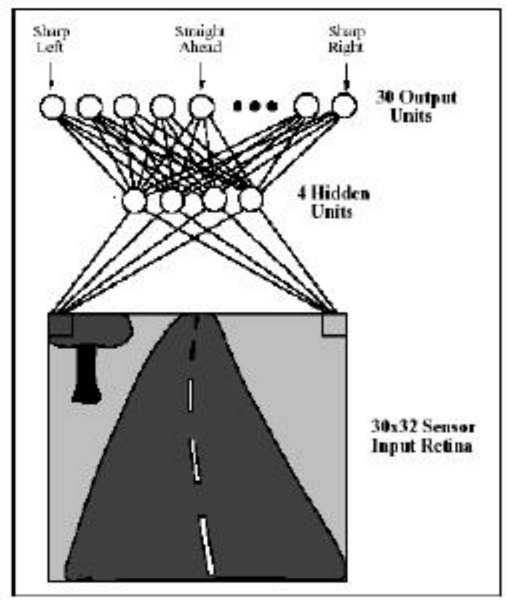
例如：互联网搜索



机器学习技术正在支撑着各种搜索引擎

例如：自动汽车驾驶（即将改变人类生活）





美国在20世纪80年代就开始研究基于机器学习的汽车自动驾驶技术

机器学习能做什么？

小数据上就已经很有用

例如：画作鉴别 （艺术）

画作鉴别(painting authentication)：确定作品的真伪

出自 [J. Hughes et al., PNAS 2009]

勃鲁盖尔（1525-1569）
的作品？

梵高（1853-1890）
的作品？

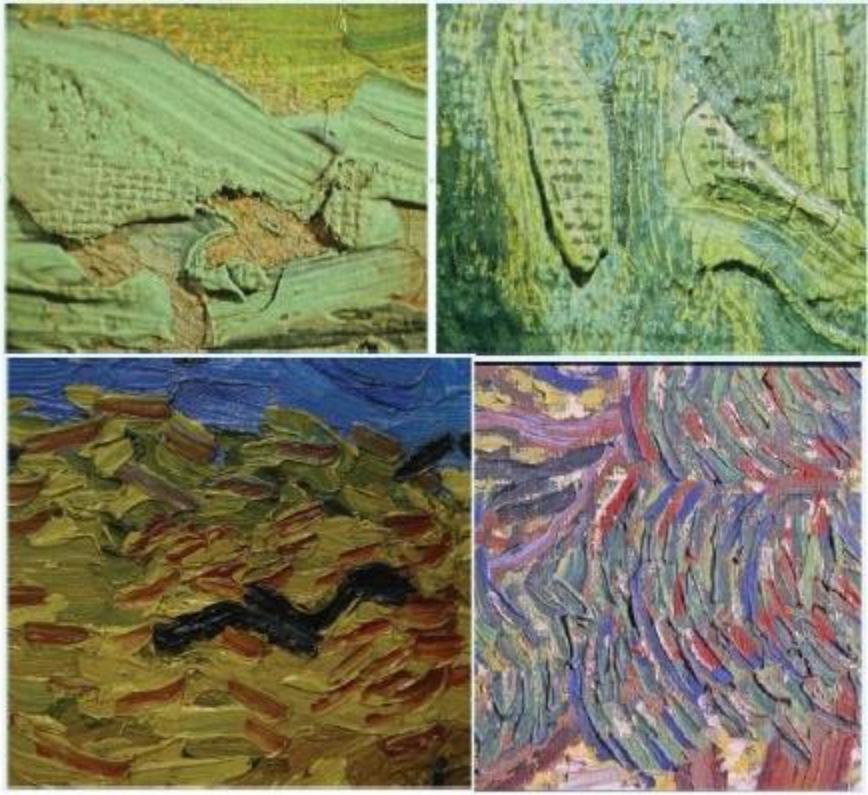
出自 [C. Johnson et al., IEEE-SP, 2008]

例如：画作鉴别 （艺术）

除专用技术手段外， 笔触分析 (brushstroke analysis) 是 画 作 鉴定的重要工具；它旨在从视觉上判断画作中是否具有艺术家的特有“笔迹”。

该工作对专业知识要求极高

- 具有较高的绘画艺术修养
- 掌握画家的特定绘画习惯



Portions of van Gogh paintings

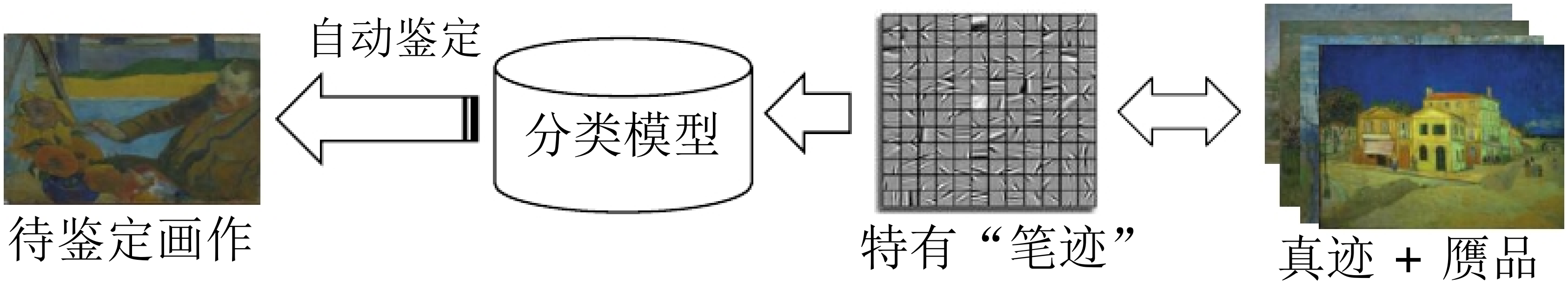
只有少数专家花费很大精力才能完成分析工作！

很难同时掌握不同时期、不同流派多位画家的绘画风格！

[C. Johnson et al., IEEE-SP, 2008]

例如：画作鉴别 （艺术）

为了降低分析成本, 机器学习技术被引入



Kröller Müller美术馆与Cornell等大学的学者对82幅梵高真迹和6幅赝品进行分析，自动鉴别精度达 **95%** [C. Johnson et al., IEEE-SP, 2008]

Dartmouth学院、巴黎高师的学者对8幅勃鲁盖尔真迹和5幅赝品进行分析，自动鉴别精度达 **100%** [J. Hughes et al., PNAS 2009][J. Mairal et al., PAMI’12]

(对用户要求低、准确高效、适用范围广)

例如：古文献修复 （文化）

古文献是进行历史研究的重要素材，但是其中很多损毁严重

Dead Sea Scrolls (死海古卷)

- 1947年出土
- 超过30,000个羊皮纸片段

Cairo Genizah

- 19世纪末被发现
- 超过300,000个片段
- 散布于全球多家博物馆

高水平专家的大量精力被用于古文献修复

[L. Wolf et al., IJCV 2011]

例如：古文献修复 （文化）

一个重要问题：

原书籍已经变成分散且混杂的多个书页，如何拼接相邻的书页？

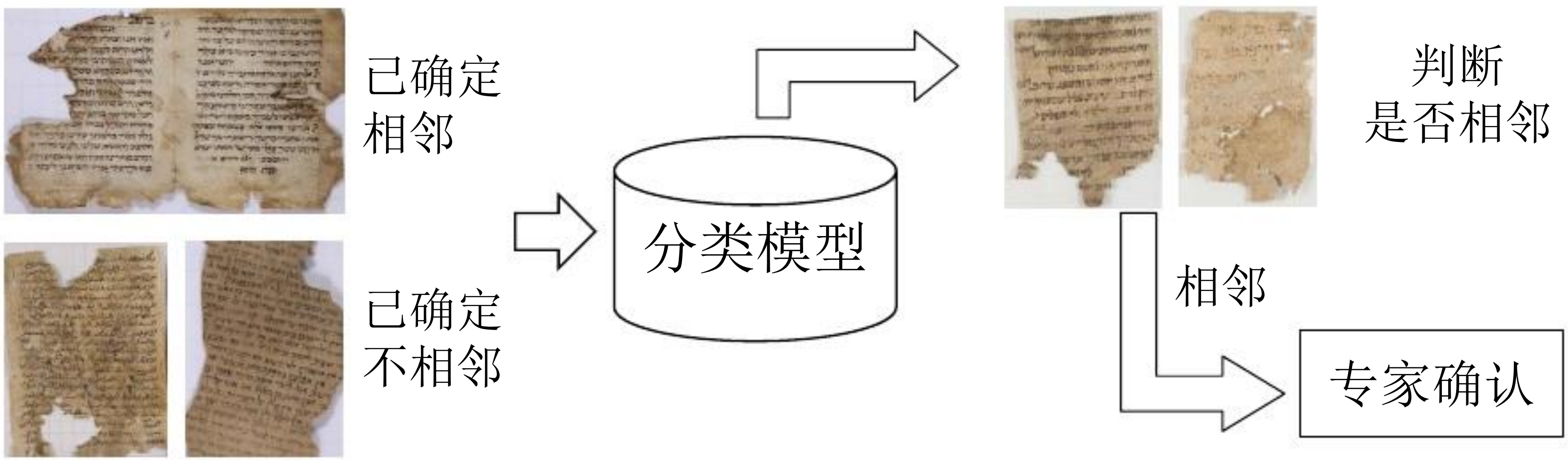
人工完成书页拼接十分困难

- 书页数量大，且分布在多处
- 部分损毁较严重，字迹模糊
- 需要大量掌握古文字的专业人才

近年来，古文献的数字化浪潮给自动文学修复提供了机会

例如：古文献修复 （文化）

以色列特拉维夫大学的学者将机器学习用于自动的书页拼接



在Cairo Genizah测试数据上，系统的自动判断精度超过 **93%**

新完成约 1,000 篇Cairo Genizah文章的拼接

(对比：过去整个世纪，数百人类专家只完成了几千篇文章拼接)

机器学习能做什么？

大数据上更惊人

例如：帮助奥巴马胜选 （政治）

How Obama's data crunchers helped him win

《时代》周刊

TIME

By **Michael Scherer**

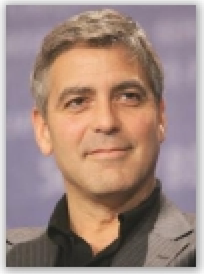
November 8, 2012 -- Updated 1645 GMT (0045 HKT) | Filed under: **Web**

例如：帮助奥巴马胜选 （政治）

通过机器学习模型：

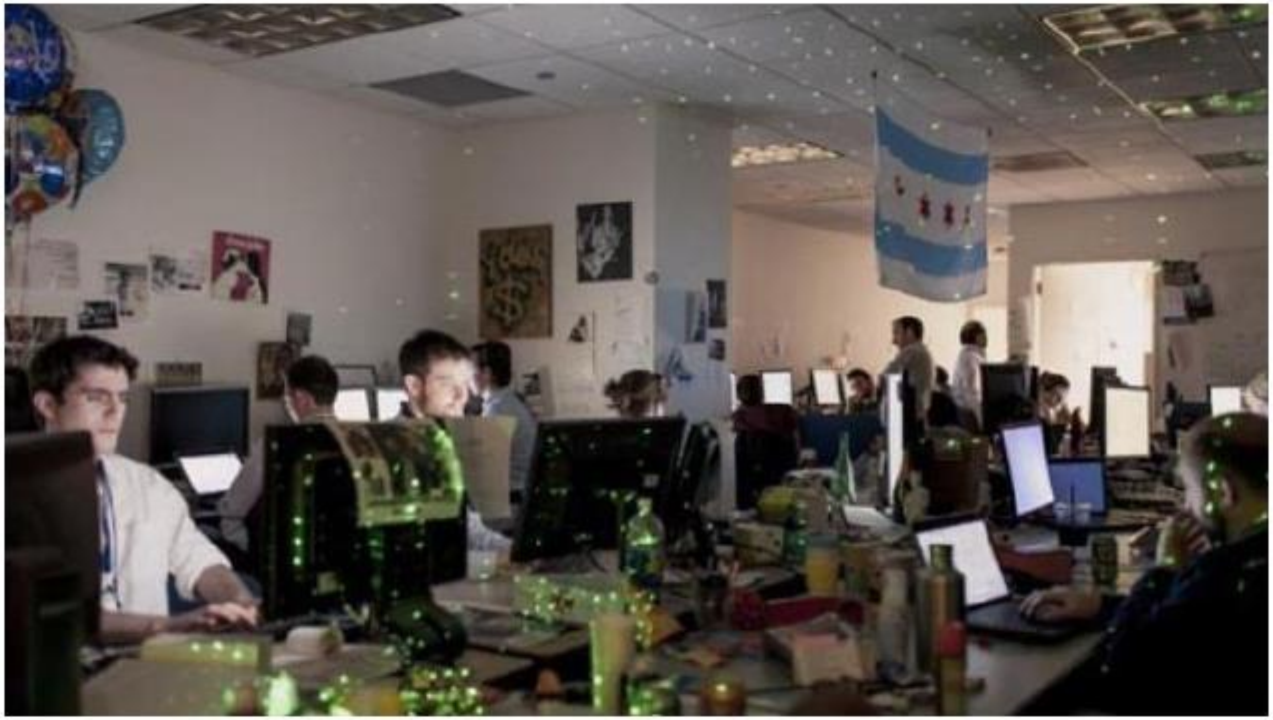
- ◆ 在总统候选人第一次辩论后，分析出哪些选民将倒戈，为每位选民找出一个最能说服他的理由
- ◆ 精准定位不同选民群体，建议购买冷门广告时段，广告资金效率比2008年提高14%
- ◆ 向奥巴马推荐，竞选后期应当在什么地方展开活动 —— 那里有很多争取对象
- ◆ 借助模型帮助奥巴马筹集到创纪录的10亿美元

例如：利用模型分析出，明星乔治克鲁尼（George Clooney）对于年龄在40-49岁的美西地区女性颇具吸引力，而她们恰是最愿意为和克鲁尼/奥巴马共进晚餐而掏钱的人 …… 乔治克鲁尼为奥巴马举办的竞选筹资晚宴成功募集到1500万美元



- ◆ … …

例如：帮助奥巴马胜选 （政治）



队长：Rayid Ghani

卡内基梅隆大学机器学习系首任系主任Tom Mitchell教授的博士生

这个团队行动保密，定期向奥巴马报送结果；
被奥巴马公开称为总统竞选的
“核武器按钮”(“They are our nuclear codes”)

机器学习源自“人工智能”

约翰 麦卡锡
(1927-2011)
“人工智能之父”
1971年图灵奖

Artificial Intelligence (AI), 1956 -

1956年夏　　美国达特茅斯学院

J. McCarthy, M. Minsky, N. Lochester, C. E. Shannon,
H.A. Simon, A. Newell, A. L. Samuel 等10余人

达特茅斯会议标志着人工智能这一学科的诞生

John McCarthy (1927 - 2011):

1971年获图灵奖, 1985年获IJCAI终身成就奖。人工智能之父。他提出了“人工智能”的概念，设计出函数型程序设计语言Lisp，发展了递归的概念，提出常识推理和情境演算。出生于共产党家庭，从小阅读《10万个为什么》，中学时自修CalTech的数学课程，17岁进入CalTech时免修两年数学，22岁在Princeton获博士学位，37岁担任Stanford大学AI实验室主任。

第一阶段：推理期

1956-1960s: Logic Reasoning

- 出发点: “数学家真聪明！”
- 主要成就: 自动定理证明系统 (例如，西蒙与纽厄尔的“Logic Theorist”系统)

渐渐地，研究者们意识到，仅有逻辑推理能力是不够的 …

赫伯特 西蒙
(1916-2001)
1975年图灵奖

阿伦 纽厄尔
(1927-1992)
1975年图灵奖

第二阶段：知识期

爱德华 费根鲍姆
(1936-)
1994年图灵奖

1970s -1980s: Knowledge Engineering

- 出发点: “知识就是力量！”
- 主要成就: 专家系统 (例如，费根鲍姆等人的“DENDRAL”系统)

渐渐地，研究者们发现，要总结出知识再“教”给系统，实在太难了 ...

第三阶段：学习期

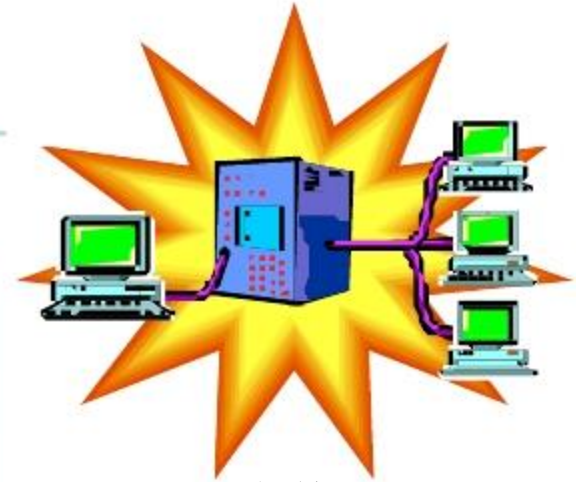
1990s -now: Machine Learning

- 出发点: “让系统自己学！”
- 主要成就: ......

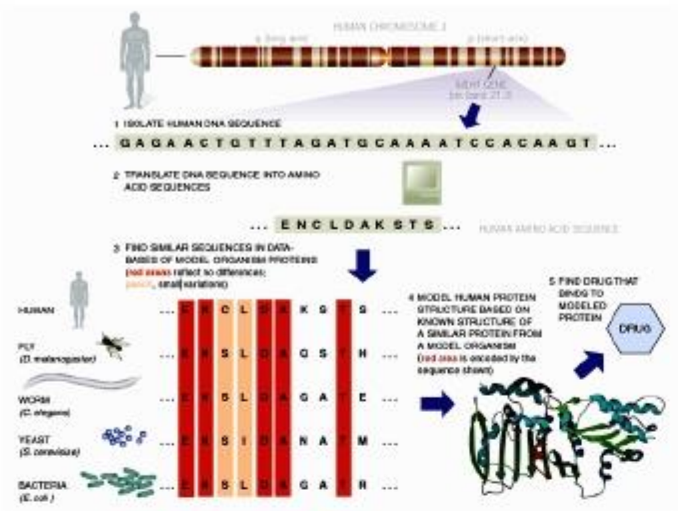
机器学习是作为“突破知识工程瓶颈”之利器而出现的

恰好在20世纪90年代中后期，人类发现自己淹没在数据的汪洋中，对自动数据分析技术——机器学习的需求日益迫切

机器学习已经“无处不在”



入侵检测



生物信息学

Web搜索

决策助手(DARPA)

汽车自动驾驶
(DARPA Grand Challenge)

火星机器人 (JPL)

大数据时代的关键技术

奥巴马提出“大数据计划”后，美国NSF进一步加强资助UC Berkeley研究如何整合将”数据”转变为”信息”的三大关键技术——机器学习、云计算、众包(crowd sourcing)

National Science Foundation: In addition to funding the Big Data solicitation, and keeping with its focus on basic research, NSF is implementing a comprehensive, long-term strategy that includes new methods to derive knowledge from data; infrastructure to manage, curate, and serve data to communities; and new approaches to education and workforce development. Specifically, NSF is:

- Encouraging research universities to develop interdisciplinary graduate programs to prepare the next
- Funding ... 10 million Expeditions in Computing project based at the University of California, Berkeley, that will integrate three powerful approaches for turning data into information - machine learning, cloud computing, and crowd sourcing;
- Providing the first round of grants to support “EarthCube” – a system that will allow geoscientists to access, analyze and share information about our planet;
- Issuing a $2 million award for a research training group to support training for undergraduates to use graphical and visualization techniques for complex data.
- Providing $1.4 million in support for a focused research group of statisticians and biologists to determine protein structures and biological pathways.
- Convening researchers across disciplines to determine how Big Data can transform teaching and learning.

整合三大关键技术

大数据时代，机器学习必不可少

收集、传输、存储大数据的目的，

是为了“利用”大数据

没有机器学习技术分析大数据，

“利用”无从谈起

小故事："机器学习"名字的由来

1952 年，阿瑟·萨缪尔 (Arthur Samuel, 1901—1990) 在 IBM 公司研制了一个西洋跳棋程序，这个程序具有自学习能力，可通过对大量棋局的分析逐渐辨识出当前局面下的"好棋"和"坏棋"，从而不断提高弈棋水平，并很快就下赢了萨缪尔自己. 1956 年，萨缪尔应约翰·麦卡锡 (John McCarthy, "人工智能之父", 1971 年图灵奖得主) 之邀，在标志着人工智能学科诞生的达特茅斯会议上介绍这项工作. 萨缪尔发明了"机器学习"这个词，将其定义为"不显式编程地赋予计算机能力的研究领域". 他的文章 "Some studies in machine learning using the game of checkers" 1959 年在 *IBM Journal* 正式发表后，爱德华·费根鲍姆 (Edward Feigenbaum, "知识工程之父", 1994 年图灵奖得主) 为编写其巨著 *Computers and Thought*，在 1961 年邀请萨缪尔提供一个该程序最好的对弈实例. 于是，萨缪尔借机向康涅狄格州的跳棋冠军、当时全美排名第四的棋手发起了挑战，结果萨缪尔程序获胜，在当时引起轰动.

事实上，萨缪尔跳棋程序不仅在人工智能领域产生了重大影响，还影响到整个计算机科学的发展. 早期计算机科学研究认为，计算机不可能完成事先没有显式编程好的任务，而萨缪尔跳棋程序否证了这个假设. 另外，这个程序是最早在计算机上执行非数值计算任务的程序之一，其逻辑指令设计思想极大地影响了 IBM 计算机的指令集，并很快被其他计算机的设计者采用.

References

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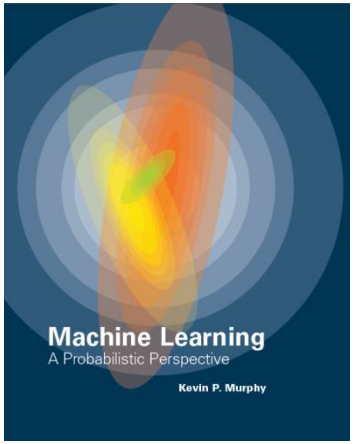
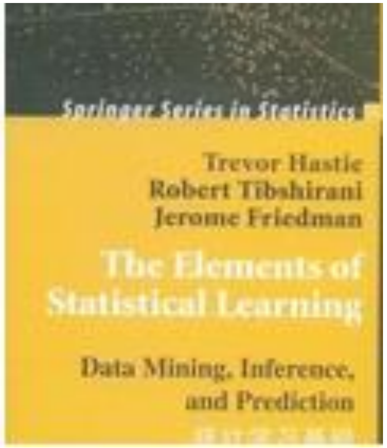
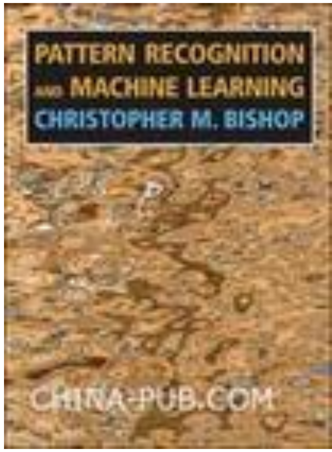
2. T. Hastie, R. Tibshirani and J. H. Friedman, Elements of Statistical Learning: Data Mining, Inference, and Prediction
3. Christopher Bishop, Pattern Recognition and Machine Learning. Springer, 2006.







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5. Richard Duda, Peter Hart and David Stork, Pattern Classification, 2nd ed. John Wiley & Sons, 2001.
6. Tom Mitchell, Machine Learning. McGraw-Hill, 1997.
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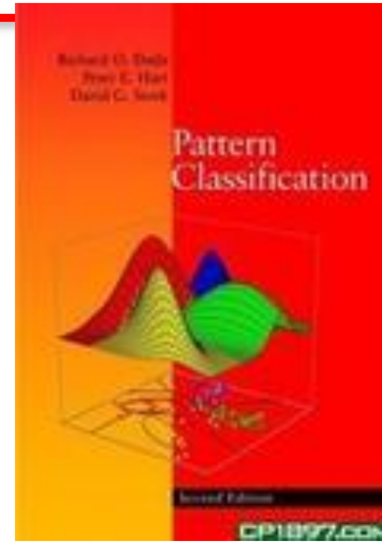
8. 李航，统计学习方法，清华大学出版社，2012.



Journals and Conferences

Machine Learning (journal)

Journal of Machine Learning Research

Neural Computation (journal)

Journal of Intelligent Systems(journal)

International Conference on Machine Learning (ICML) (conference)

Neural Information Processing Systems (NIPS) (conference)

Chapter 1: Introduction

What is Machine Learning

- Machine learning informally as the that gives computers to learn — that gives computers the ability to learn without being explicitly programmed (Arthur Samuel, 1959).
- A computer program is set to learn from an experience E with respect to some task T and some performance measure P if its performance on T as measured by P improves with experience E (Tom Mitchell, 1998).

Machine Learning History

- Machine Learning has origins in Computer Science
- Pattern Recognition has origins in Engineering
- They are different facets of the same field

Types of Machine Learning (1/2)

1. ~~Predictive or Supervised learning~~

The goal is to learn a mapping from inputs $\mathbf{x}$ to outputs y, given a labeled set of input-output pairs $D = \{(\mathbf{x}_i, y_i)\}^N_{i=1}$.

When y_i is categorical, the problem is known as **classification** or **pattern recognition**, and when y_i is real-valued, the problem is known as **regression**.

Another variant, known as **ordinal regression有序回归**, occurs where label space Y has some natural ordering, such

Types of Machine Learning (2/2)

2. Descriptive or Unsupervised learning

Given inputs, $D = \{\mathbf{x}_i\}^N_{i=1}$, and the goal is to find "interesting patterns" in the data. This is sometimes called **knowledge discovery**. This is a much less well-defined problem.

3. Reinforcement learning

This is useful for learning how to act or behave when given occasional reward or punishment signals.

Supervised Learning (1/3)

- Classification

Here the goal is to learn a mapping from inputs $\mathbf{x}$ to outputs y, where $y \in \{1, . . . , C\}$, with C being the number of classes.

If $C = 2$, this is called **binary classification** (in which case we often assume $y \in \{0, 1\}$); if $C > 2$, this is called **multiclass classification**.

If the class labels are not mutually exclusive, we call it **multi-label classification**.

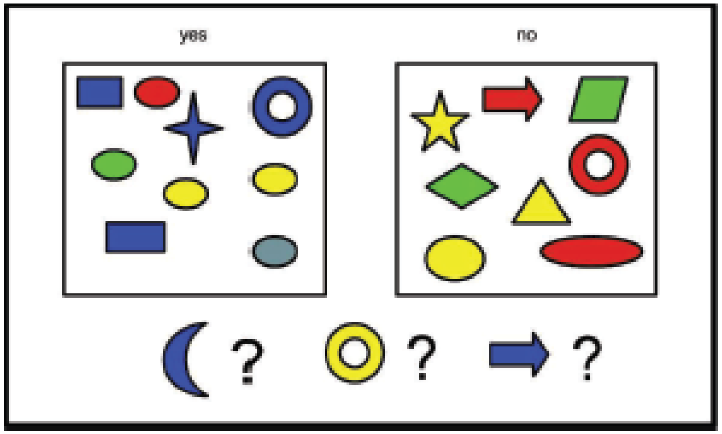
Supervised Learning (2/3)

One way to formalize the problem is as **function approximation函数逼近**.

We assume $y = f(\mathbf{x})$ for some unknown function f, and the goal of learning is to estimate the function f given a labeled training set, and then to make predi $\hat{y} = \hat{f}(\mathbf{x})$ using .

Our main goal is to make predictions on novel inputs, meaning ones that we have not seen before (this is called **generalization泛化**).

Supervised Learning (3/3)



(a)

D features (attributes)

N cases

Color	Shape	Size (cm)	Label
Blue	Square	10	1
Red	Ellipse	2.4	1
Red	Ellipse	20.7	0

(b)

Figure 1.1 Left: Some labeled training examples of colored shapes, along with 3 unlabeled test cases. Right: Representing the training data as an $N \times D$ design matrix. Row i represents the feature vector $\mathbf{x}_i$. The last column is the label, $y_i \in \{0, 1\}$. Based on a figure by Leslie Kaelbling.

Supervised Learning (1/2)

- Regression

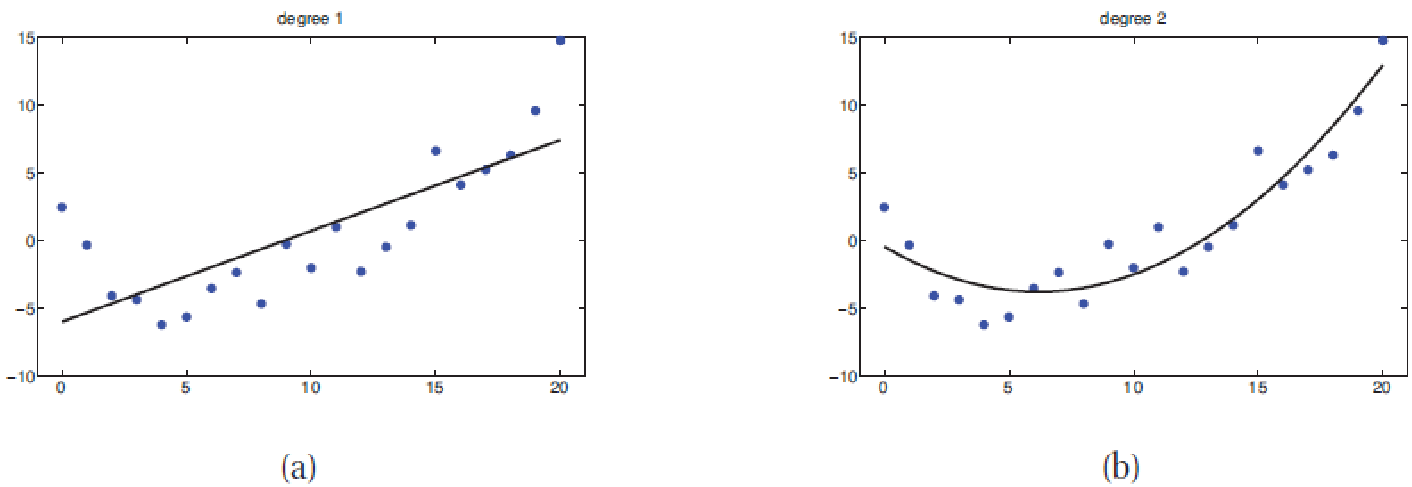


Figure 1.7 (a) Linear regression on some 1d data. (b) Same data with polynomial regression (degree 2). Figure generated by `linregPolyVsDegree`.

Supervised Learning (2/2)

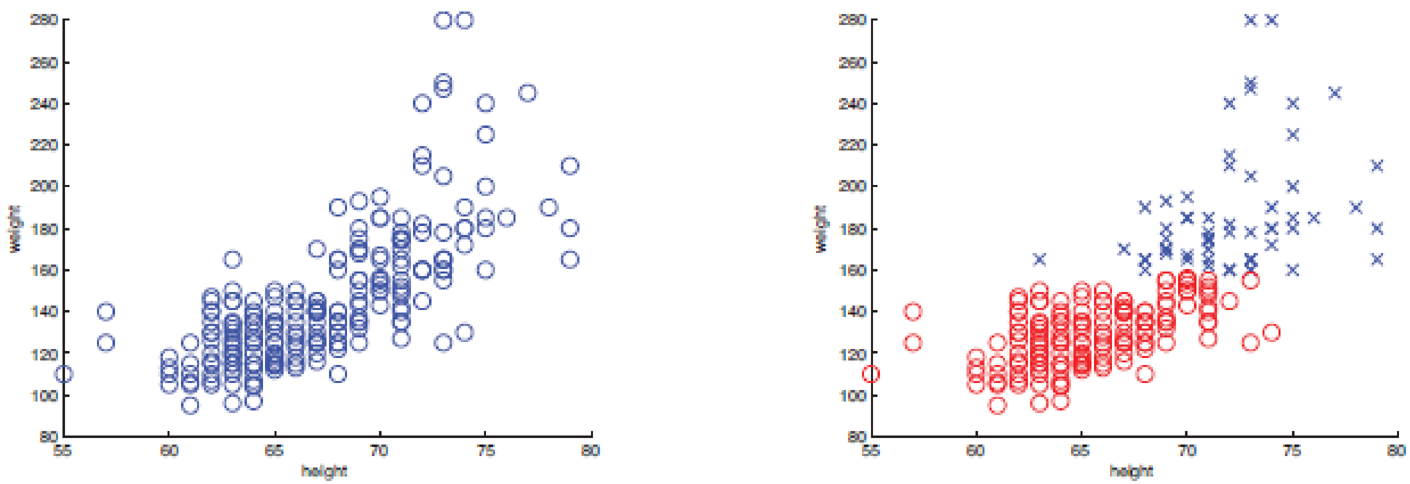
- Real-world regression problems.
  - Predict tomorrow's stock market price given current market conditions and other possible side information.
  - Predict the age of a viewer watching a given video on YouTube.
  - Predict the temperature at any location inside a building using weather data, time, door sensors, etc.

Unsupervised Learning (1/7)

- Formalize our task as one of density estimation, that is, we want to build models of the form $p(x_i|\boldsymbol{\theta})$.
  - We have written $p(x_i|\theta)$ instead of $p(y_i|x_i, \theta)$;
  - x_i is a vector of features, so we need to create multivariate probability models.

Unsupervised Learning (2/7)

- Discovering clusters
  - To estimate the distribution over the number of clusters, $p(K|D)$; K denote the number of clusters.
  - To estimate which cluster each data point belongs to.

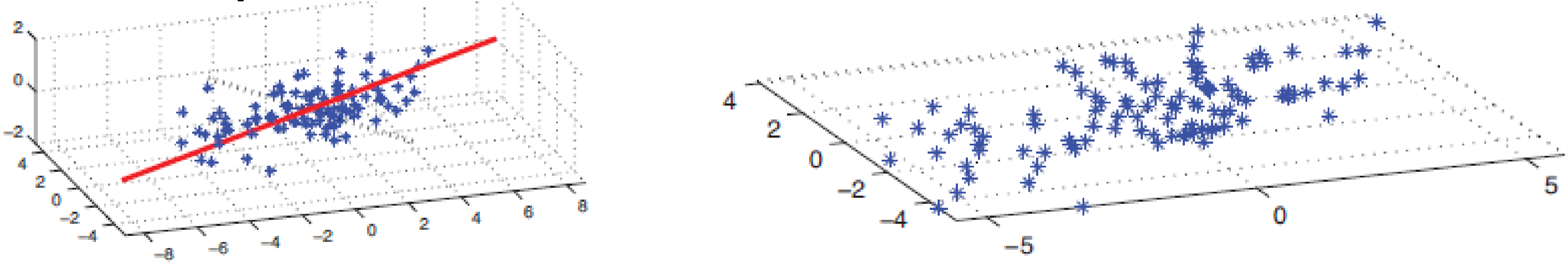


(a) (b)

Figure 1.8 (a) The height and weight of some people. (b) A possible clustering using $K = 2$ clusters. Figure generated by `kmeansHeightWeight`.

Unsupervised Learning (3/7)

- Discovering latent factors
  - Dimensionality reduction: When dealing with high dimensional data, it is often useful to reduce the dimensionality by projecting the data to a lower dimensional subspace which captures the "essence" of the data.



(a) (b)

Figure 1.9 (a) A set of points that live on a 2d linear subspace embedded in 3d. The solid red line is the first principal component direction. The dotted black line is the second PC direction. (b) 2D representation of the data. Figure generated by pcaDemo3d.

- The motivation behind this technique is that although the data may appear high dimensional, there may only be a small number of degrees of variability, corresponding to **latent factors**.

Figure 1.10 a) 25 randomly chosen 64×64 pixel images from the Olivetti face database. (b) The mean and the first three principal component basis vectors (eigenfaces). Figure generated by `pcaImageDemo`.

Unsupervised Learning (5/7)

- Discovering graph structure
  - Measure a set of correlated variables to discover which ones are most correlated with which others.
  - This can be represented by a graph G, in which nodes represent variables, and edges represent direct dependence between variables.

$$\hat{G} = \operatorname{argmax} p(G|\mathcal{D})$$

Unsupervised Learning (6/7)

- Matrix completion

Sometimes we have missing data, that is, variables whose values are unknown. The corresponding design matrix will then have "holes" in it. The goal of **imputation** is to infer plausible values for the missing entries.

Unsupervised Learning (7/7)

- Collaborative filtering:

A common example of this concerns predicting which movies people will want to watch based on how they, and other people, have rated movies which they have already seen.

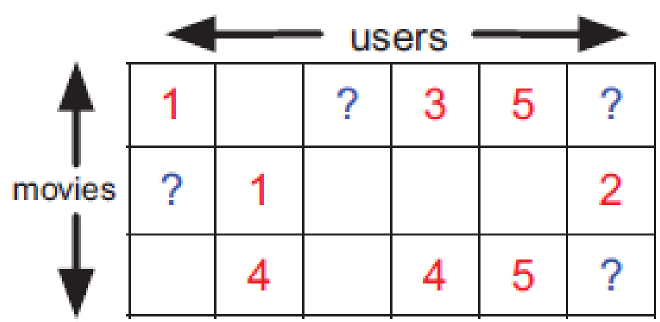
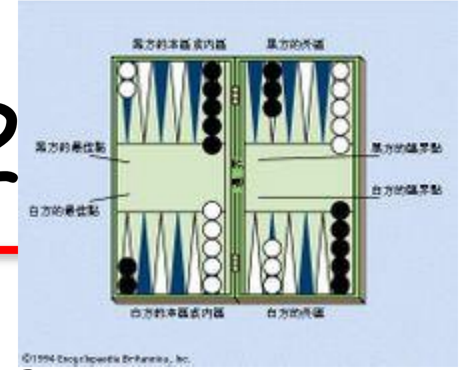


Figure 1.13 Example of movie-rating data. Training data is in red, test data is denoted by ?, empty cells are unknown.

Reinforcement learning (1/2)

- *Reinforcement learning*（强化学习）(Sutton and Barto, 1998): finding suitable actions to take in a given situation in order to maximize a reward.
- Here the learning algorithm is not given examples of optimal outputs, in contrast to supervised learning, but must instead discover them by a process of trial and error.

Reinforcement learning (2/2)

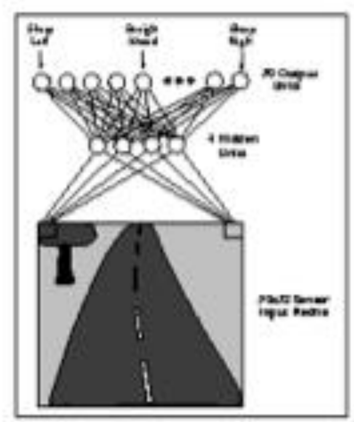


- A general feature of reinforcement learning is the trade-off between *exploration*探索, in which the system tries out new kinds of actions to see how effective they are, and *exploitation*利用, in which the system makes use of actions that are known to yield a high reward.
- Too strong a focus on either exploration or exploitation will yield poor results.

Some Successful Applications of Machine Learning (1/2)

- Learning to drive an autonomous vehicle
  - Train computer-controlled vehicles to steer correctly
  - Drive at 70 mph for 90 miles on public highways
  - Associate steering commands with image sequences

ALVINN [Pomerleau] drives 70 mph on highways



Some Successful Applications of Machine Learning (2/2)

- Handwriting Recognition
  - Task T– recognizing and classifying handwritten words within images
  - Performance measure P – percent of words correctly classified
  - Training experience E– a database of handwritten words with given classifications

The Machine Learning Approach

Data Collection

Samples

Model Selection

Probability distribution to model process

Parameter Estimation

Values/distributions

Generalization

(Training)

Search

Find optimal solution to problem

Decision

(Inference

OR

Testing)

Example 1(1/11)

- A fish packing plant wants to automate the process of sorting incoming fish on a conveyor belt according to species.
- Set up a camera, take some sample images and begin to note some physical differences between the two types of fish — length, lightness, width, number and shape of fins, position of the mouth, and so on.

Example 1 (2/11)

- First the camera captures an image of the fish.
- The camera's signals are preprocessed to simplify subsequent operations without loosing relevant information.
- Segmentation operation: the image of different fish are somehow isolated from one another and from the background.

Example 1 (3/11)

- Feature extraction: to reduce the data by measuring certain "features" or "properties."
- These features are then passed to a classifier that evaluates the evidence presented and makes a final decision as to the species.

Example 1 (4/11)

Salmon

Sea bass

Example 1 (5/11)

Example 1 (6/11): Representation: Fish Length As Feature

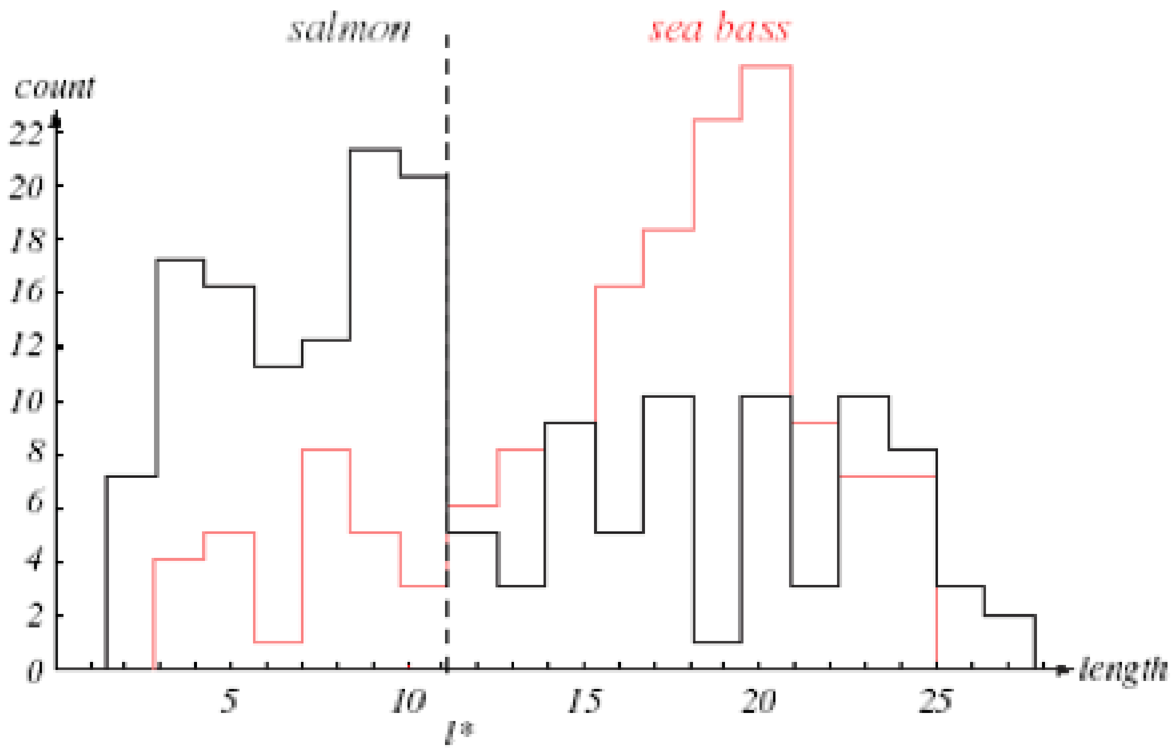


FIGURE 1.2. Histograms for the length feature for the two categories. No single threshold value of the length will serve to unambiguously discriminate between the two categories; using length alone, we will have some errors. The value marked l^* will lead to the smallest number of errors, on average. From: Richard O. Duda, Peter E. Hart, and David G. Stork, *Pattern Classification*. Copyright © 2001 by John Wiley & Sons, Inc.

Training (design or learning) Samples

Example 1 (7/11): Fish Lightness As Feature

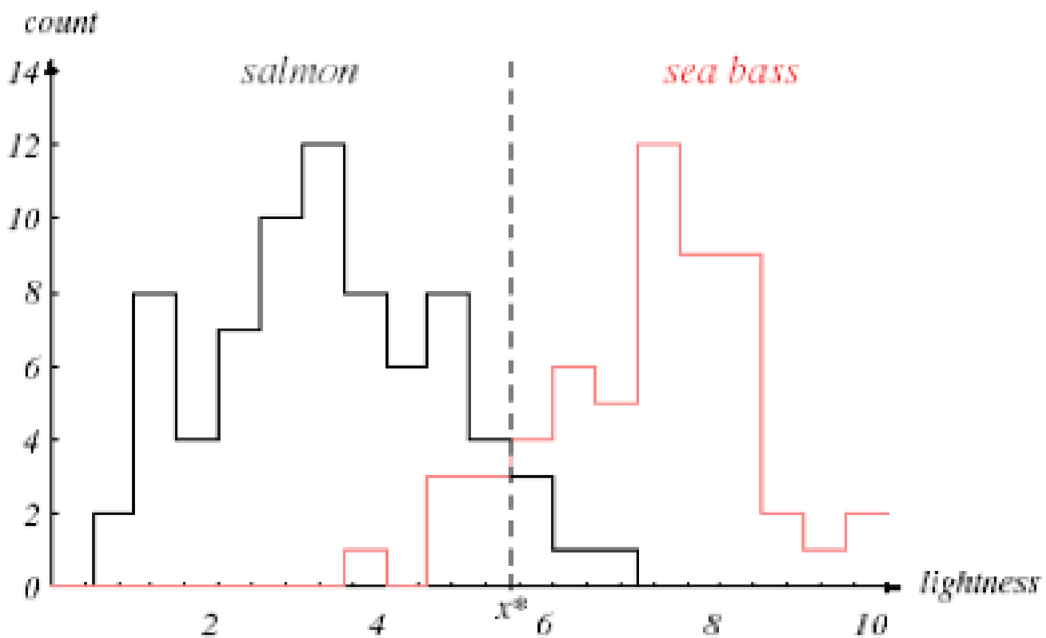
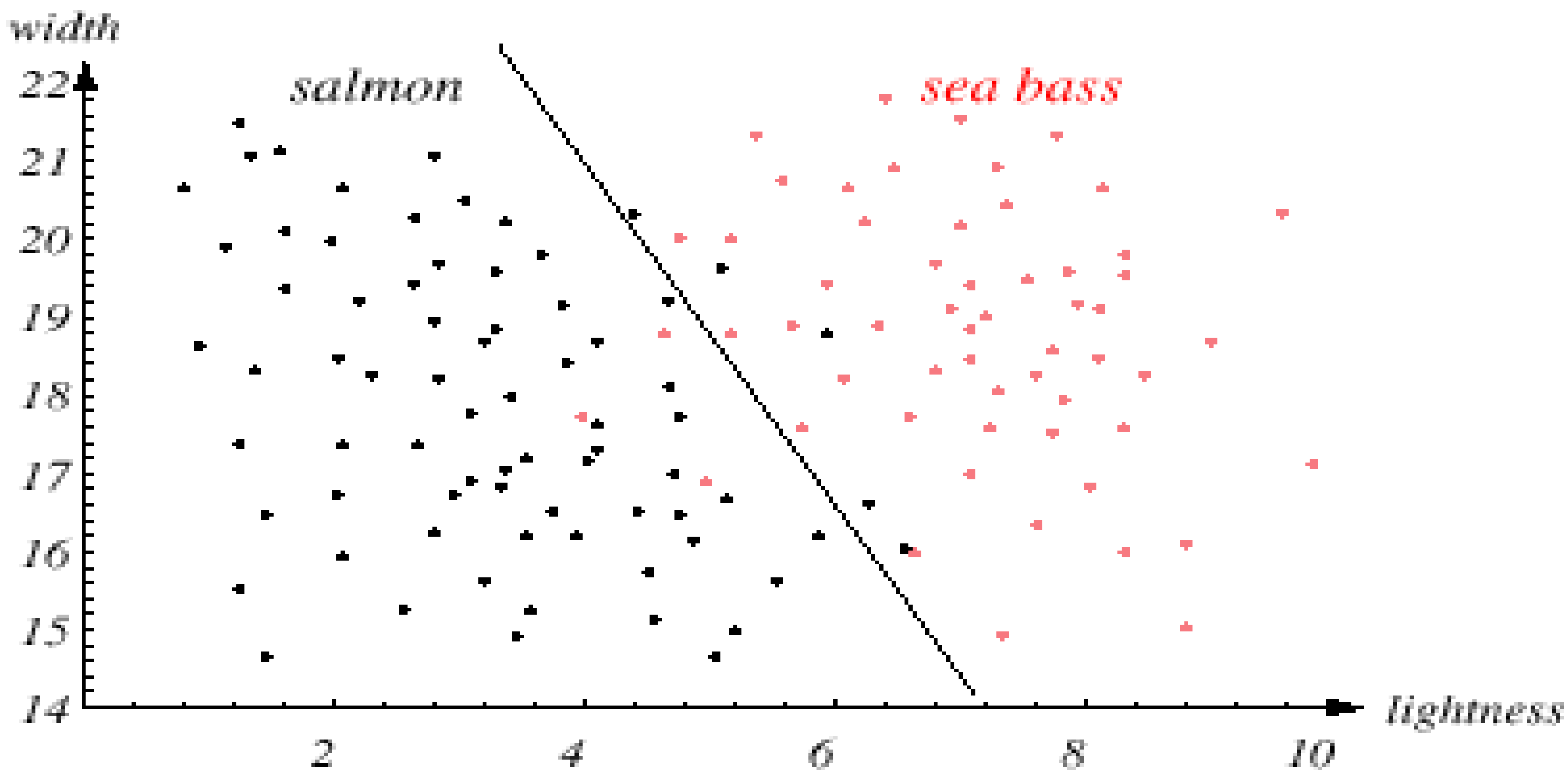


FIGURE 1.3. Histograms for the lightness feature for the two categories. No single threshold value x^* (decision boundary) will serve to unambiguously discriminate between the two categories; using lightness alone, we will have some errors. The value x^* marked will lead to the smallest number of errors, on average. From: Richard O. Duda, Peter E. Hart, and David G. Stork, *Pattern Classification*. Copyright © 2001 by John Wiley & Sons, Inc.

Overlap of these histograms is small compared to length feature

Example 1 (8/11)



Example 1 (9/11): Two-dimensional Feature Space

Linear (simple) decision boundary; Cost of misclassification?

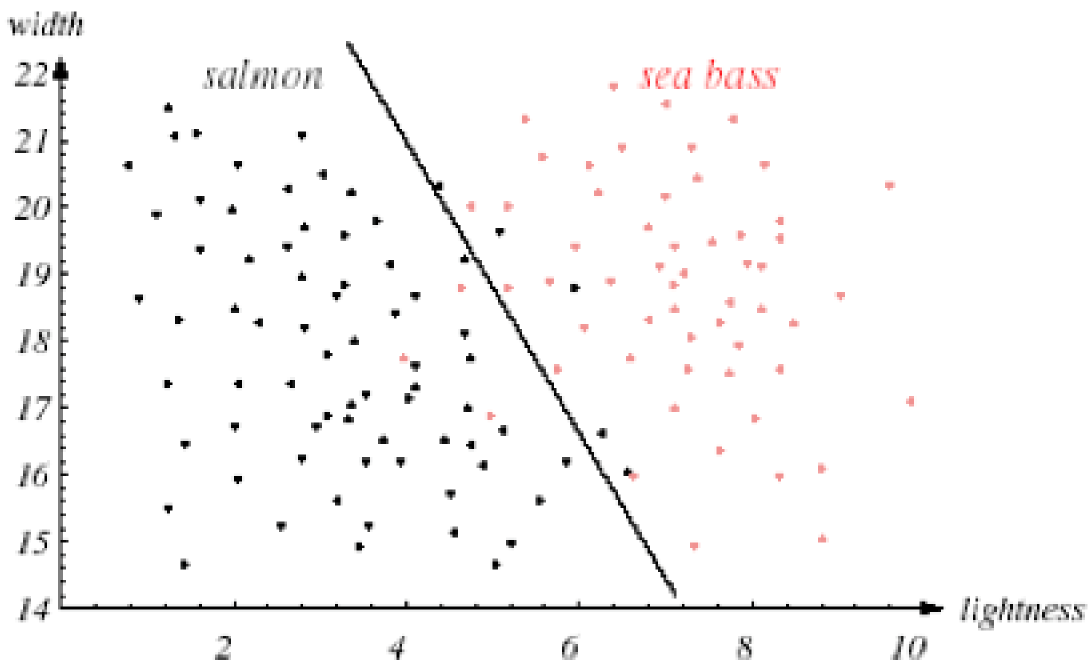


FIGURE 1.4. The two features of lightness and width for sea bass and salmon. The dark line could serve as a decision boundary of our classifier. Overall classification error on the data shown is lower than if we use only one feature as in Fig. 1.3, but there will still be some errors. From: Richard O. Duda, Peter E. Hart, and David G. Stork, *Pattern Classification*. Copyright © 2001 by John Wiley & Sons, Inc.

Two features together are better than individual features

Example 1 (10/11): Complex Decision Boundary

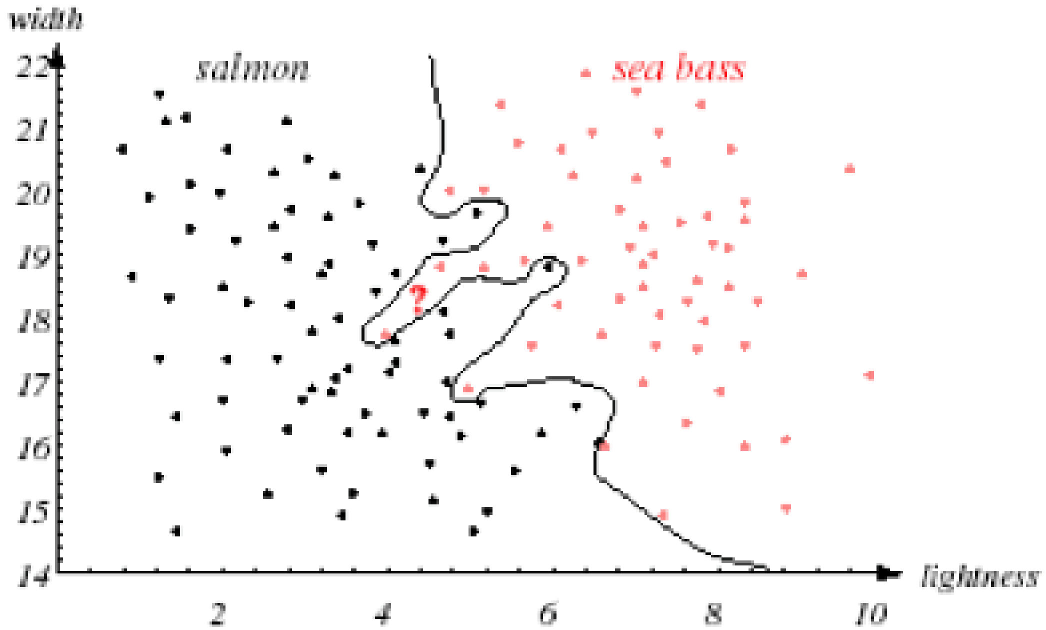


FIGURE 1.5. Overly complex models for the fish will lead to decision boundaries that are complicated. While such a decision may lead to perfect classification of our training samples, it would lead to poor performance on future patterns. The novel test point marked ? is evidently most likely a salmon, whereas the complex decision boundary shown leads it to be classified as a sea bass. From: Richard O. Duda, Peter E. Hart, and David G. Stork, *Pattern Classification*. Copyright © 2001 by John Wiley & Sons, Inc.

Generalization ability of the learned boundary

Example 1 (11/11): Boundary With Good Generalization

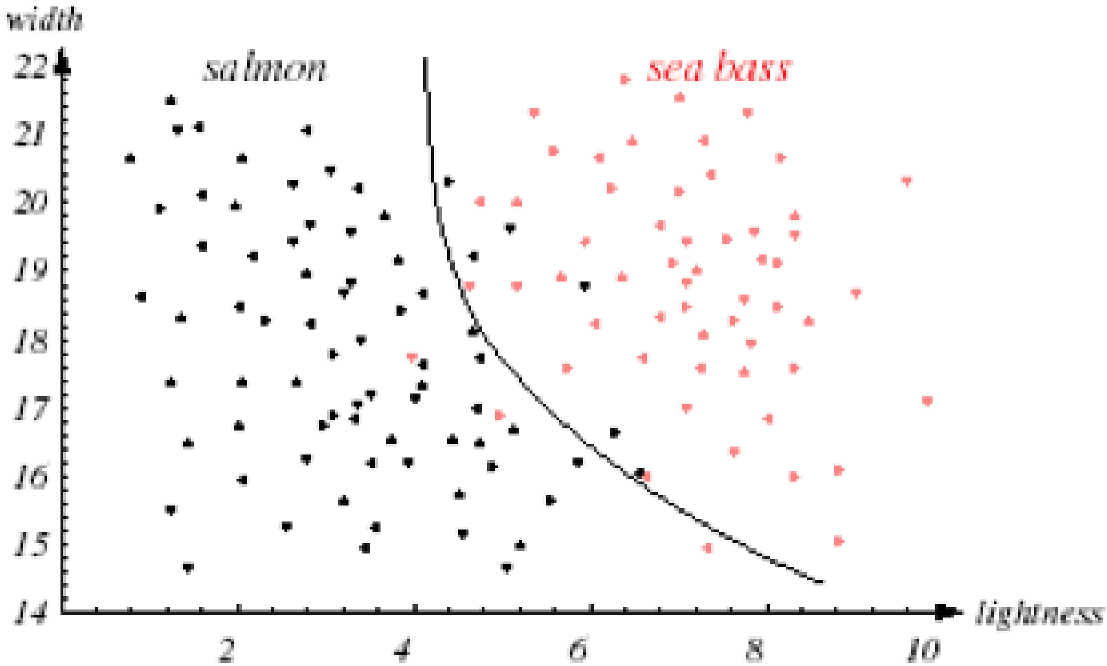
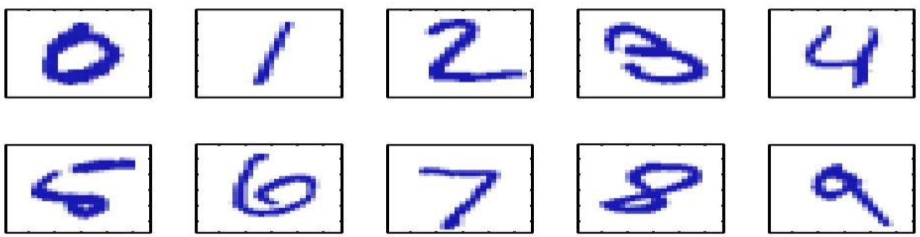


FIGURE 1.6. The decision boundary shown might represent the optimal tradeoff between performance on the training set and simplicity of classifier, thereby giving the highest accuracy on new patterns. From: Richard O. Duda, Peter E. Hart, and David G. Stork, *Pattern Classification*. Copyright © 2001 by John Wiley & Sons, Inc.

Simple decision boundaries are preferred

Example 2 (1/4)

Handwritten Digit Recognition



Wide variability of same numeral

- Handcrafted rules will result in large number of rules and exceptions.
- Better to have a machine that learns from a large training set

Example 2 (2/4)

- A large set of N digits $\{x_1,\ldots,x_N\}$ called a *training set* is used to tune the parameters of an adaptive model.
- We can express the category of a digit using *target vector t*, which represents the identity of the corresponding digit.
- Note that there is one such target vector t for each digit image x.

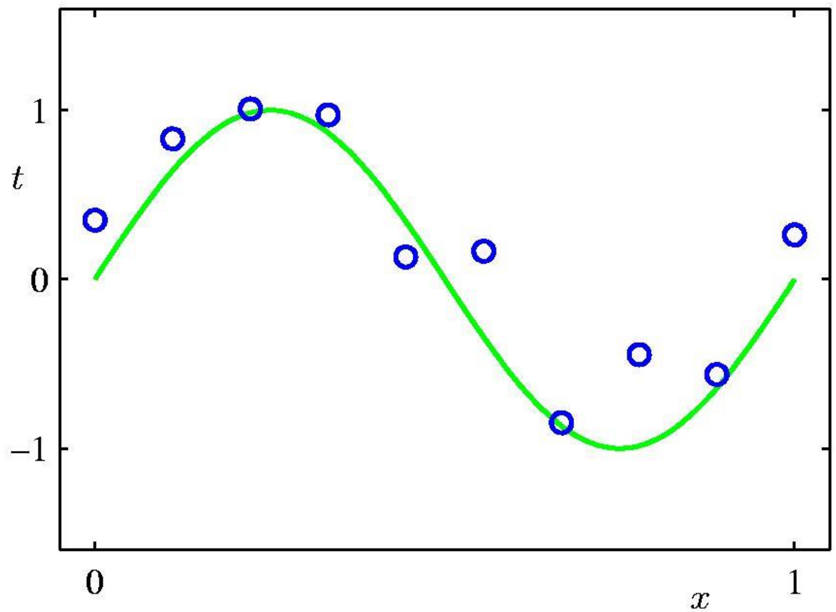
Example 2 (3/4)

- The result of running the machine learning algorithm can be expressed as a function $y(x)$.
- The precise form of the function $y(x)$ is determined during the *training phase*, also known as the *learning phase*, on the basis of the *training data*.
- Once the model is trained it can then determine the identity of new digit images, which are said to comprise a *test set*.
- The ability to categorize correctly new examples that differ from those used for training is known as *generalization*.

Example 2 (4/4)

- For most practical applications, the original input variables are typically *preprocessed* to transform them into some new space of variables
- This pre-processing stage is sometimes also called *feature extraction.*
- Note that new test data must be pre-processed using the same steps as the training data.

Polynomial Curve Fitting



Plot of a training data set of N = 10 points, shown as blue circles, each comprising an observation of the input variable x along with the corresponding target variable t. The green curve shows the function sin(2πx) used to generate the data.

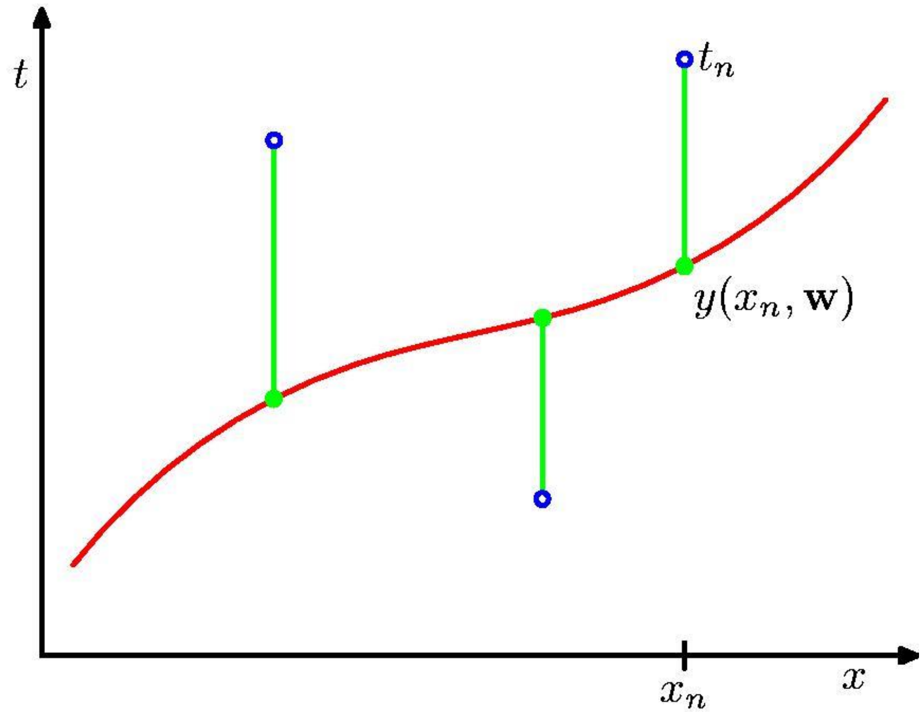
Our goal is to predict the value of t for some new value of x, without knowledge of the green curve.

- Our goal is to exploit this training set in order to make predictions of the value t^ of the target variable for some new value x^ of the input variable.
- Where M is the order of the polynomial, and x^j denotes x raised to the power of j. The polynomial coefficients $w_0,\ldots,w_M$ are collectively denoted by the vector w.

$$y(x, \mathbf{w}) = w_0 + w_1 x + w_2 x^2 + \ldots + w_M x^M = \sum_{j=0}^{M} w_j x^j$$

- Functions, such as the polynomial, which are linear in the unknown parameters have important properties and are called linear models.

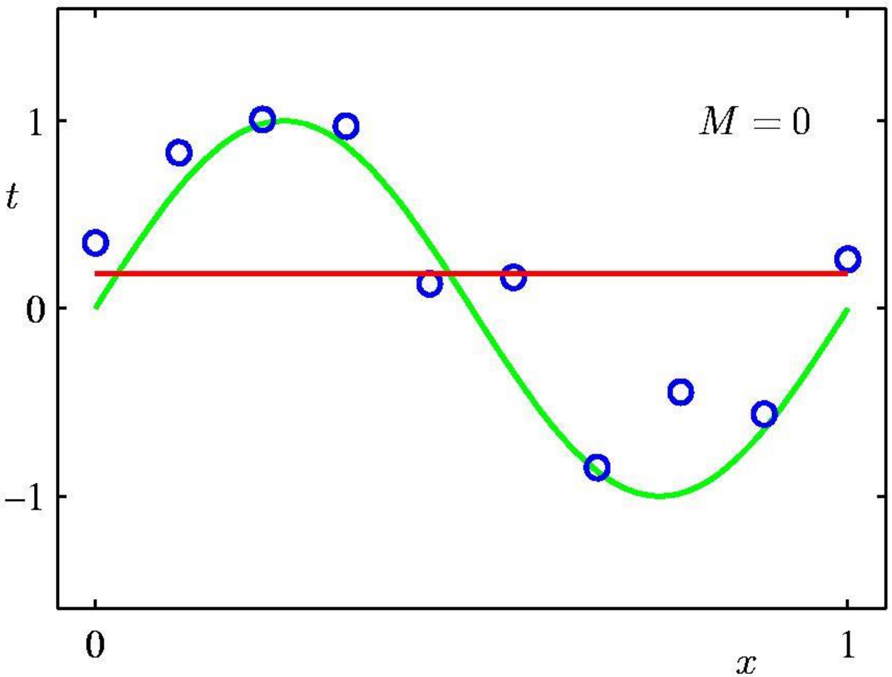
Sum-of-Squares Error Function



The error function corre- sponds to (one half of) the sum of the squares of the displacements (shown by the vertical green bars) of each data point from the function y(x, w).

$$E(\mathbf{w}) = \frac{1}{2}\sum_{n=1}^{N}\{y(x_n, \mathbf{w}) - t_n\}^2$$

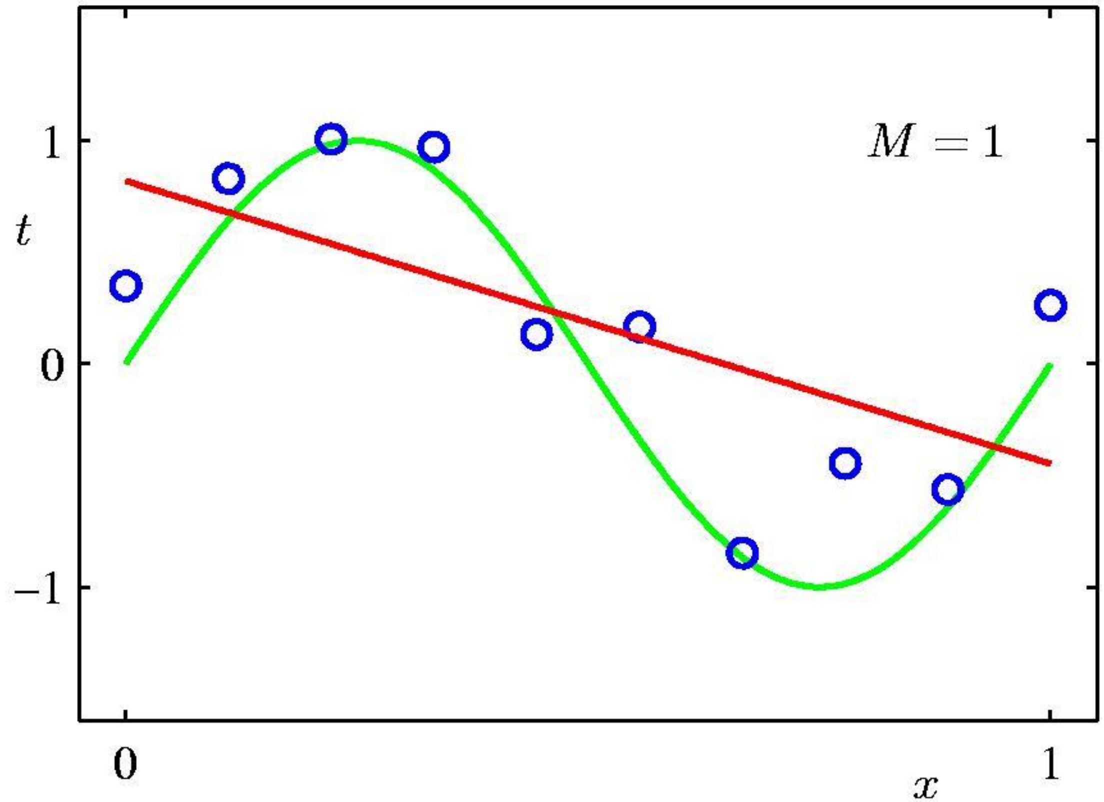
0th Order Polynomial



•Model comparison or model selection: there remains the problem of choosing the order M of the polynomial.

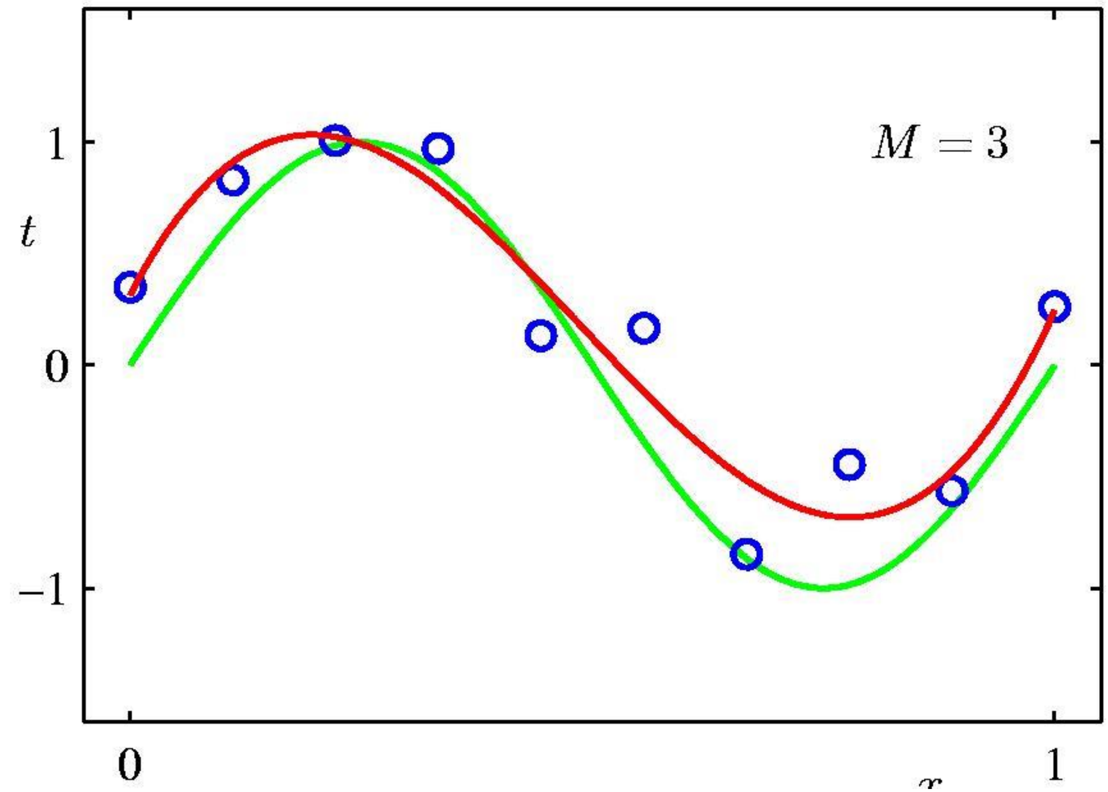
$$y(x, \mathbf{w}) = w_0 + w_1 x + w_2 x^2 + \ldots + w_M x^M = \sum_{j=0}^{M} w_j x^j$$

1^{st} Order Polynomial



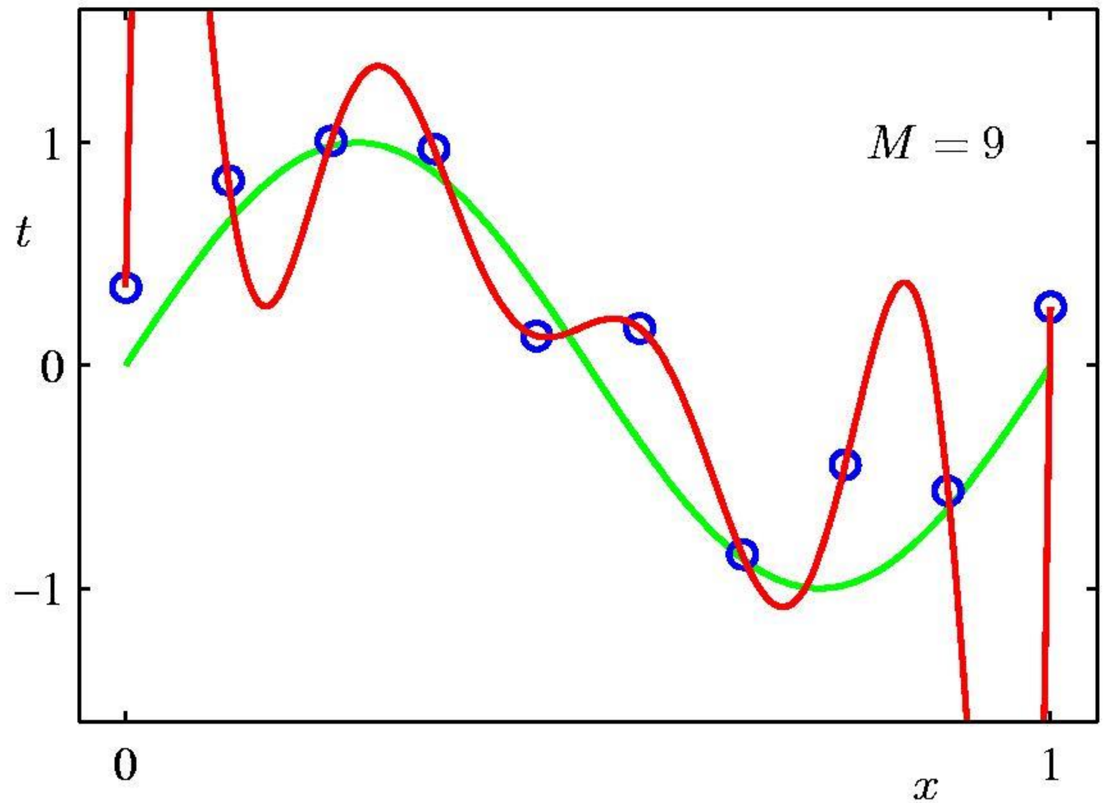
$$y(x, \mathbf{w}) = w_0 + w_1 x + w_2 x^2 + \ldots + w_M x^M = \sum_{j=0}^{M} w_j x^j$$

3rd Order Polynomial



$$y(x, \mathbf{w}) = w_0 + w_1 x + w_2 x^2 + \ldots + w_M x^M = \sum_{j=0}^{M} w_j x^j$$

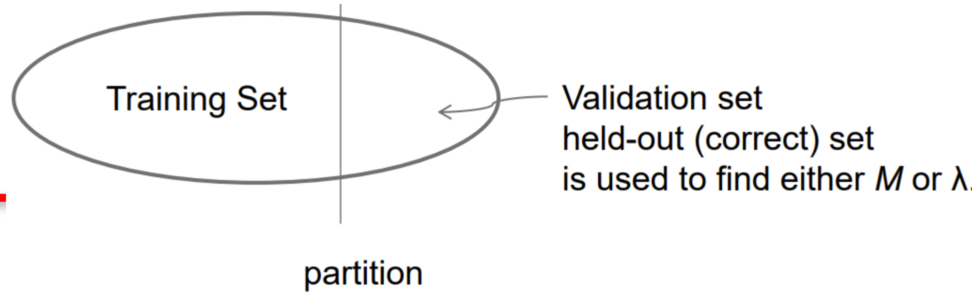
9^{th} Order Polynomial

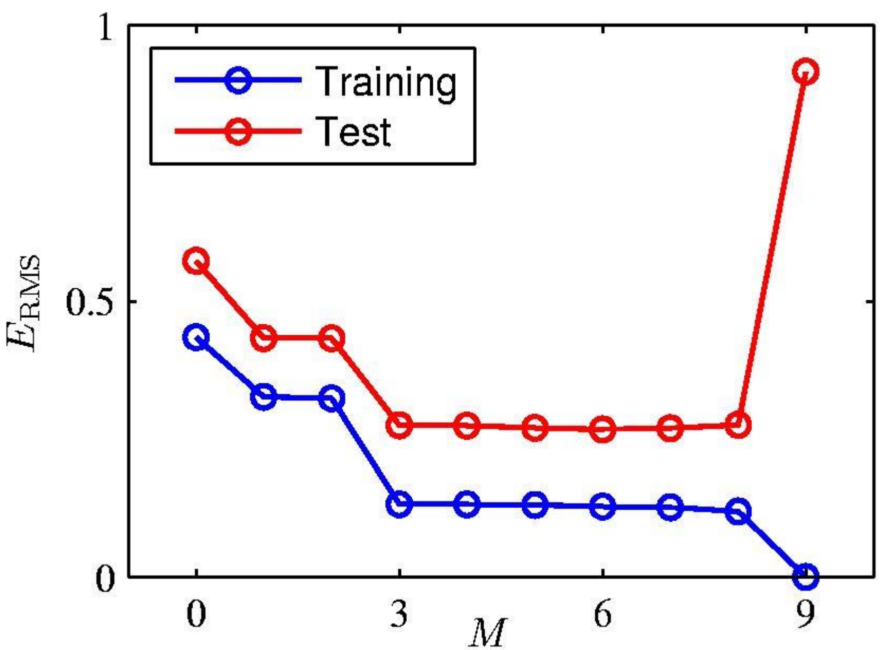


$$y(x, \mathbf{w}) = w_0 + w_1 x + w_2 x^2 + \ldots + w_M x^M = \sum_{j=0}^{M} w_j x^j$$

$$E(\mathbf{w}) = \frac{1}{2} \sum_{n=1}^{N} \{y(x_n, \mathbf{w}) - t_n\}^2$$

Over-fitting





Graphs of the root-mean-square error, defined by the followed formula, evaluated on the training set and on an independent test set for various values of M.

Root-Mean-Square (RMS) Error: $E_{\mathrm{RMS}} = \sqrt{2E(\mathbf{w}^\star)/N}$

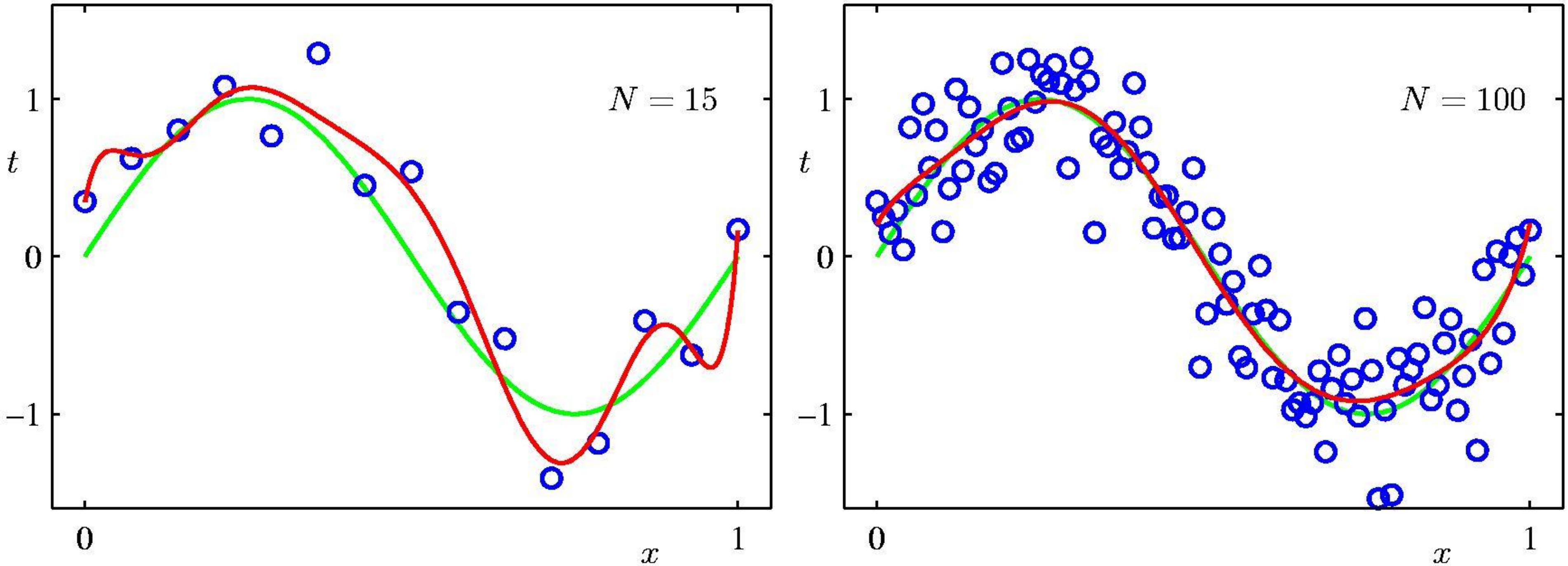
Polynomial Coefficients

	$M = 0$	$M = 1$	$M = 3$	$M = 9$
$w_0^\star$	0.19	0.82	0.31	0.35
$w_1^\star$		-1.27	7.99	232.37
$w_2^\star$			-25.43	-5321.83
$w_3^\star$			17.37	48568.31
$w_4^\star$				-231639.30
$w_5^\star$				640042.26
$w_6^\star$				-1061800.52
$w_7^\star$				1042400.18
$w_8^\star$				-557682.99
$w_9^\star$				125201.43

Table of the coefficients w for polynomials of various order. Observe how the typical magnitude of the coefficients increases dramatically as the order of the polynomial increases.

Data Set Size:

9th Order Polynomial



Plots of the solutions obtained by minimizing the sum-of-squares error function using the M = 9 polynomial for N = 15 data points (left plot) and N = 100 data points (right plot). <u>We see that increasing the size of the data set reduces the over-fitting</u> problem.

Regularization

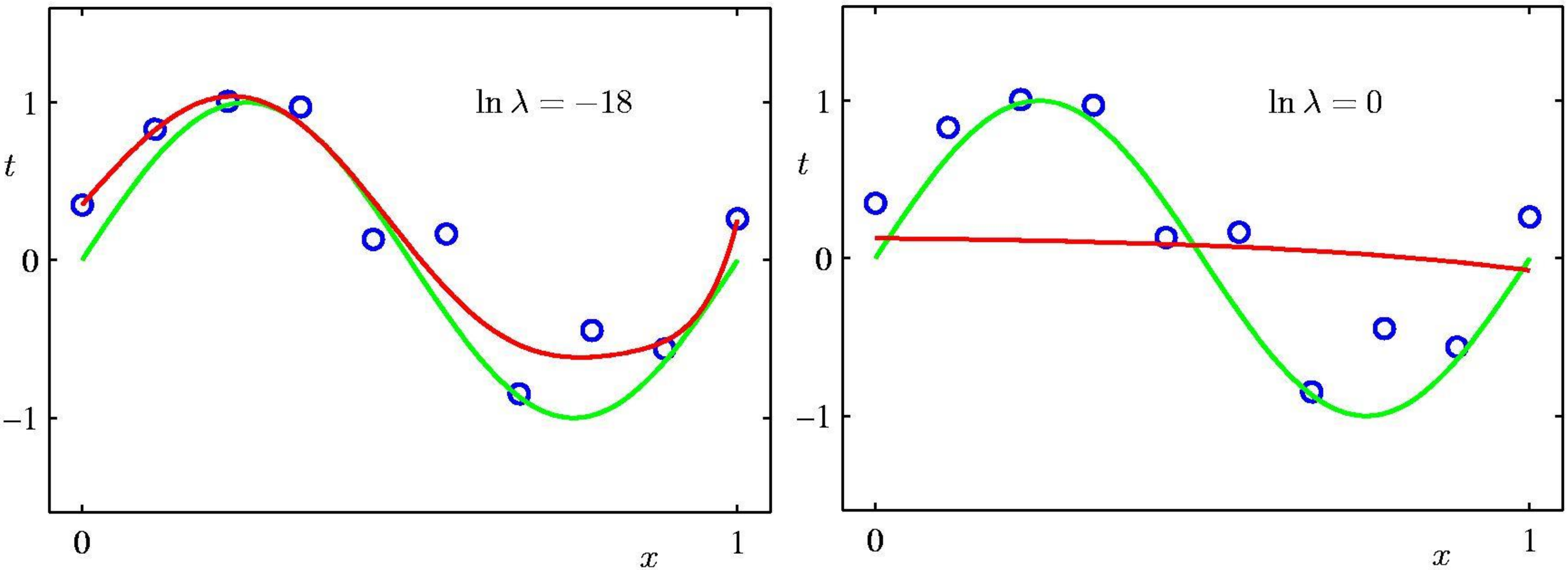
Penalize large coefficient values

$$\widetilde{E}(\mathbf{w}) = \frac{1}{2}\sum_{n=1}^{N}\{y(x_n, \mathbf{w}) - t_n\}^2 + \frac{\lambda}{2}\|\mathbf{w}\|^2$$

$$\|\mathbf{w}\|^2 \equiv \mathbf{w}^{\mathrm{T}}\mathbf{w} = w_0^2 + w_1^2 + \ldots + w_M^2$$

the coefficient λ governs the relative importance of the regularization term compared with the sum-of-squares error term.

Regularization:



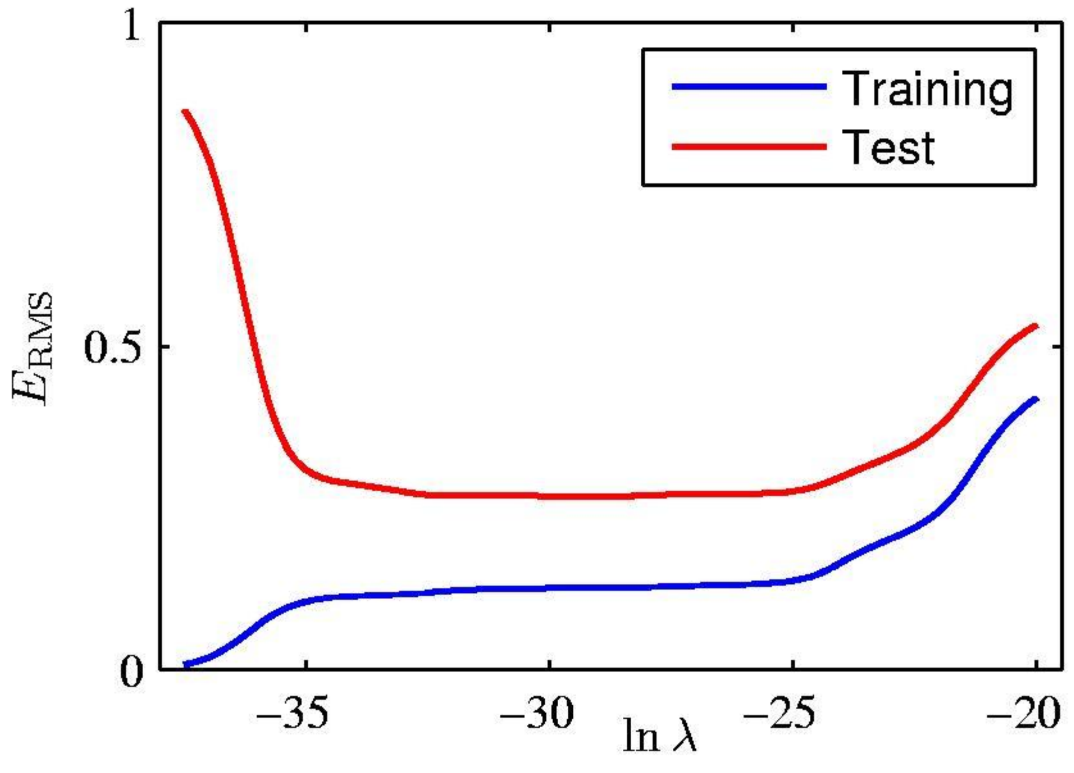
Plots of M = 9 polynomials fitted to the data set shown in Figure using the regularized error function for two values of the regularization parameter λ corresponding to lnλ = −18 and lnλ = 0. The case of no regularizer, i.e., λ = 0, corresponding to lnλ = −∞, is shown at previous Figure.

Polynomial Coefficients

	$\ln \lambda = -\infty$	$\ln \lambda = -18$	$\ln \lambda = 0$
$w_0^\star$	0.35	0.35	0.13
$w_1^\star$	232.37	4.74	-0.05
$w_2^\star$	-5321.83	-0.77	-0.06
$w_3^\star$	48568.31	-31.97	-0.05
$w_4^\star$	-231639.30	-3.89	-0.03
$w_5^\star$	640042.26	55.28	-0.02
$w_6^\star$	-1061800.52	41.32	-0.01
$w_7^\star$	1042400.18	-45.95	-0.00
$w_8^\star$	-557682.99	-91.53	0.00
$w_9^\star$	125201.43	72.68	0.01

Table of the coefficients w for M = 9 polynomials with various values for the regularization parameter λ. Note that lnλ = −∞ corresponds to a model with no ~~regularization. We see that, as the value of λ increases, the typical magnitude of~~ the coefficients gets smaller.

Regularization: E_{RMS} vs. $\ln \lambda$



Graph of the root-mean-square error versus lnλ for the M = 9 polynomial.

Thank You!